PPS3MVCD

DATASHEET

PRODUCT SPECIFICATION

3/4" CMOS monochrome VGA event-based sensor

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REVISION HISTORY

Release date	Version	Changes
10 – January – 2020	1.0	Initial version.
21 – February – 2020	1.1	- Fixed typos. - Updated package pinlist.
24 – April – 2020	1.2	- Fixed typos, missing references and consistency of nomenclature across tables. - Vdda restricted to ±5%. - Updated pinlist. - Removed external trigger. - Added glossary section. - Added QE from LIFO. - Updated die thickness after packaging.
03 – August – 2020	1.3	- Updated pinlist. - Added definition of asserted TPD1. - Added MSL qualification.
01 – February – 2021	1.4	- Added SPI pictures. - Included comment on soldering profile.
09 – March- 2021	1.5	- Updated package pin list table. - Updated PCB schematic. - Updated global illuminance section. - Updated self-test control section. - Replace SPI pin name SSn by CSn
08 – March – 2022	1.6	- Removed T&C KC link

APPLICATIONS

- High speed counting
- Vibration monitoring
- Kinematic monitoring

FEATURES

- 15 µm x 15 µm pixel
- Optical format 3/4”
- Typical latency 200µs
- Dynamic range >120dB

DESCRIPTION

PPS3MVCD is a VGA event-based vision sensor developed around 15-um x 15-um contrast detection pixels.

This vision sensor technology features low power consumption, always-on full awareness, high dynamic range and compressed data output.

A high level of programmability is guaranteed via a four-wire serial peripheral interface. It allows the user to define region of interest and configure the sensor’s operating mode.

The vision sensor data output is fully digital and consists of 21-wire parallel data bus with an external clock and data valid flag. This interface can easily be connected to an external parallel to LVDS serializer IC.

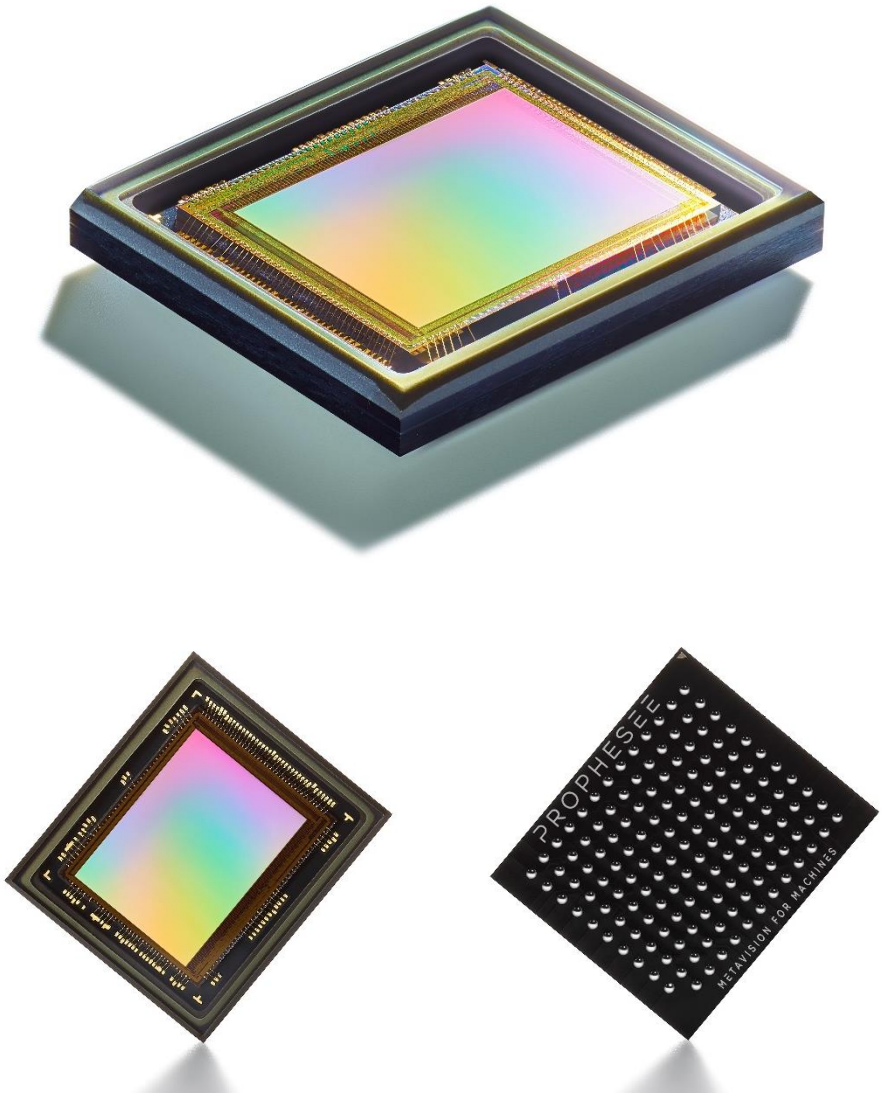

Figure 1: Packaged die picture

GLOSSARY

Term	Definition	Meaning
AER	Address-Event Representation	Transmission protocol for event-based data. An "event" is a data packet containing the pixel's address in the array plus a payload, e.g. contrast detection polarity.
Biases		Set of camera settings.
CD	Contrast Detection	Indicates "events" triggered by temporal contrast (relative intensity change). CD events have a polarity, negative or positive, depending on if triggered by decreasing or increasing light.
Event		Sensor-generated data packet containing the pixel's address in the array plus a payload, e.g. contrast detection polarity.
EVT	Event Type Definition	The Event-Based Processing Systems designed by Prophesee gather data from the sensor and encodes these data in a proprietary event stream format. This event format has been specified to strike a balance between compactness and robustness, as well as constraints imposed by the processing platforms.
ROI	Region of Interest	Part of the sensor still active while other pixels are kept in reset, to focus on specific parts of the scene.
TD	Temporal Difference	Deprecated name for CD.

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1 Functional Block Diagram

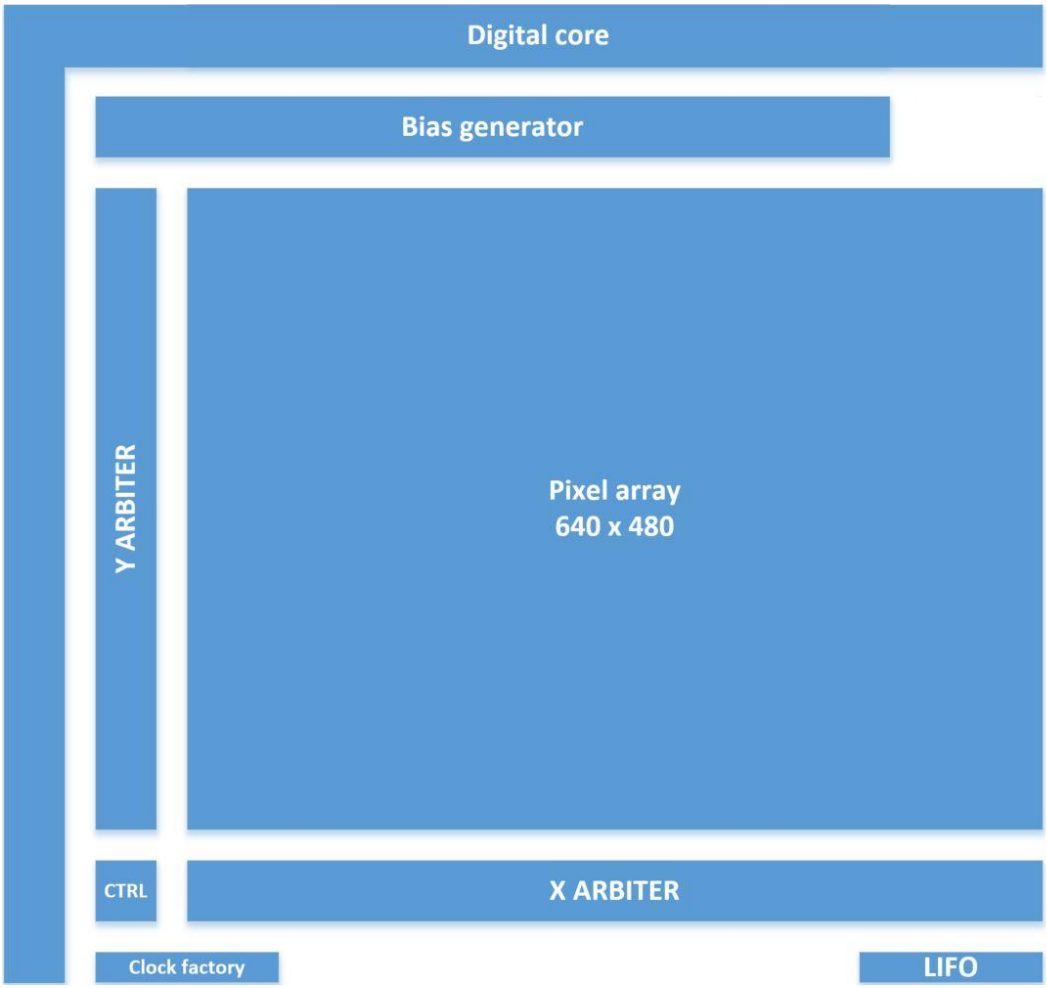


Figure 2: Functional block diagram

2 Sensor Overview

Event-based contrast-detection vision sensors constitute a new paradigm for acquiring dynamic visual information. The main advantages of these vision sensors, compared to frame-based image acquisition techniques, are low-latency response, wide-dynamic range operation and pixel-level data redundancy suppression. This frame-free operation directly reduces data volume at the sensor output. The immediate benefits of temporal redundancy suppression are reductions in bandwidth, memory and computing power requirements for data transmission and post-processing.

The in-pixel change detector is built around a fast continuous-time logarithmic photoreceptor with asynchronous event-driven signal processing. It continuously monitors photocurrent for changes and responds with polarity-discriminated events that represent a relative increase or decrease in light intensity that exceeds tunable thresholds. The occurrence of these events is sensed by one of two voltage comparators. The circuit responds to temporal contrasts over more than 6 decades of illumination (> 120 dB). In the example shown in Figure 3, the typical response of the change detector to a signal waveforms is explained. The upper trace represents arbitrary light input to a pixel and the trace below, the voltage waveform at the node tracking the photocurrent. The event generation circuitry responds with pulse events of positive or negative polarity respectively to positive and negative gradients of the photocurrent; the rate of change is encoded in the inter-event intervals.

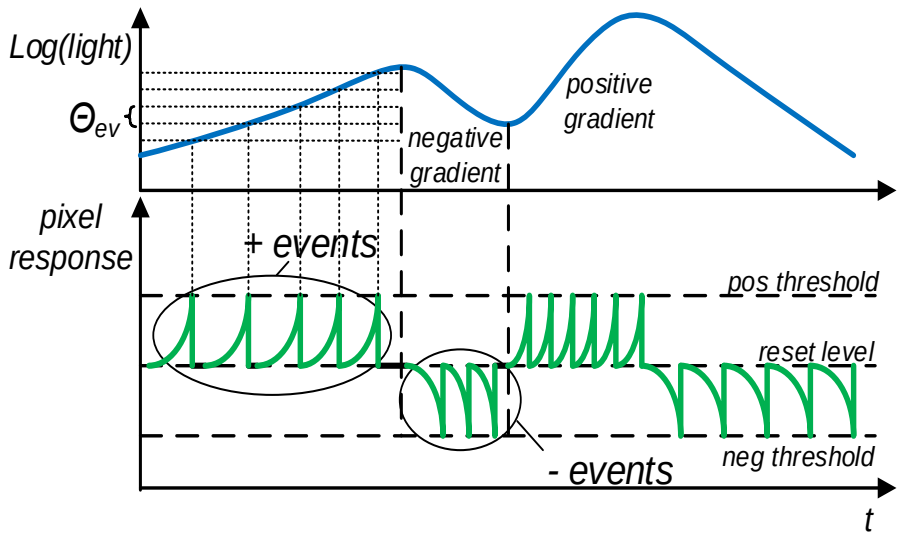


Figure 3: Typical contrast detection pixel signals

3 Absolute Maximum Ratings

Parameter	Symbol	Min	Max	Units	Justification
Absolute rating for 1.8-V supplies	ABS	-0.5	2	V	Standard 0.18um technology node
Storage temperature range	Ts	-40	125	°C	Standard packaging technology
Electrostatic discharge (ESD) Human Body Model (HBM): JS-001-2017	HBM	2000		V	Standard performance of ESD protection diode in the design
Electrostatic *discharge* (ESD) Charged Device Model (CDM): JS-002-2018	CDM	1000		V	Standard performance of ESD protection diode in the design
Latch-up: JESD-78	LU	10		mA	Contact PROPHESEE for more details
Moisture Sensitivity Level Preconditioning stress JESD22-A113	MSL	3			Allowable floor life 168 hours (7 days) at <=30/60%

Caution needs to be taken to avoid dried stains on the underside of the glass due to condensation. The glass lid glue is permeable and can absorb moisture if the sensor is placed in a high % RH environment.

Parameter	Symbol	Value	Units	Justification
Analog supply voltage	vdda	1.8 ±5%	V	
Digital supply voltage	vddd	1.8 ±10%	V	
Digital core supply voltage	vddd_core	1.8 ±10%	V	
I/O supply voltage	Vddd_io	1.8 ±10%	V	Designed to meet LVCMOS1V8

4 Electro-Optical Specifications

4.1 General Electro-Optical Specifications

Parameter	Test conditions	Value
Optical format		3/4 inch
Fabrication process		CIS 180nm
Active vision sensor size		H: 9.6 mm | V: 7.2 mm | diagonal: 12 mm
Number of active pixels		640H x 480V (PPS3MVCD)
Pixel size		15 um x 15 um (PPS3MVCD)

4.2 CD Electro-Optical Specifications (standard user settings[1], nominal power supply voltage and temperature[2])

Values below are listed for one single sensor setup (bias set) and one choice of user settings. Different trade-offs can be chosen to optimize for a given sensor specification depending on application requirements and conditions (e.g. higher speed, lower BG activity, higher contrast sensitivity, …). Please contact PROPHESEE for further details on sensor performance adaptation.

Parameter	Test conditions / comment	Min	Typ	Max	Unit
Dynamic Range DR_{CD}			>120		dB
Nominal Contrast Threshold	Standard user settings		25		%
Low-light Cut-Off (LLCO)			0.08		lux
Response Latency (max)	Typ. contrast, @LLCO		7700		µs
Response Latency (typ)	Typ. contrast, @10lux		200		µs
Response Latency (fast)	Typ. contrast, >1000lux		100		µs
Maximum readout throughput			50		Mevents/s
Max background (BG) rate	Per pixel		9		events/s
Low BG activity range	BG rate < 0.5Hz/pixel		2 - 2000		lux
$SNR_{CD,min-max}$	Post-layout simulation (low-light cut-off – high-light cut-off)		21-39		dB
Contrast Thresh Non-Uniformity			3.7		%

[1] Standard user settings are defined by the register initialization sequence. PROPHESEE cannot guarantee sensor behavior and performances if these register values are updated. Detailed definitions of parameters and test conditions are given in the PROPHESEE Knowledge Center Sensor Characterization page.

[2] Typical values from measurements at nominal power supply 1.8 V and at room temperature: 25C.

5 Electrical specifications

Power supply specifications below are listed for full performances of the vision sensor. Low power mode fully guarantees all-time awareness of the vision sensor at reduced performances. Operating power consumption can be lowered by trading off some performance parameters. Please contact PROPHESEE for further details on power optimization.

5.1 Power supplies

Parameter	Symbol	Min	Typ	Max	Units
Supply voltage, Analog	Vdda	1.71	1.8	1.89	V
Static current consumption, Analog	Idda		10		mA
Dynamic current consumption, Analog	Idda_dyn		80		pA/event
Supply voltage, Digital	Vddd	1.62	1.8	1.98	V
Static current consumption, Digital	Iddd		0.1		mA
Dynamic current consumption, Digital	Iddd_dyn		1.32		nA/event
Supply voltage, Digital core	Vddc	1.62	1.8	1.98	V
Static current consumption, Digital core	Iddc		3.7		mA
Dynamic current consumption, Digital core	Iddc_dyn		40		pA/event
Supply voltage, Digital IO	Vddio	1.62	1.8	1.98	V
Static current consumption, Digital IO	Iddio		0.2		mA
Dynamic current consumption, Digital IO	Iddio_dyn		230		pA/event

5.2 Total chip power

Parameter	Symbol	Min	Typ	Max	Units
Static power consumption (power/pixel)	P_{stat}		85		nW
Dynamic current consumption (power/event)	P_{RO}		3		nW/event
Sensor static power consumption	P_{tot}		26		mW

6 IO characteristics[3]

Digital output: pads ERROR, MISO, DY8…DY0, DX9…DX0, DV, DPOL, CLKOUT						
Parameter	**Symbol**	**Conditions**	**Min**	**Typ**	**Max**	**Units**

[3] Specifications according to digital I/O library.

LOW-level output voltage	VOL	VDDIO = min IOL = -2 mA			0.45	V
HIGH level output voltage	VOH	VDDIO = min IOH = +2 mA	VDDIO-0.45			V

Digital inputs: pads SCLK, MODI, CSn, DR, RSTn, TM, TRIGEXT						
Parameter	**Symbol**	**Conditions**	**Min**	**Typ**	**Max**	**Units**
LOW-level input voltage	VIL		-0.3		0.35*VDDIO	V
HIGH-level input voltage	VIH		0.65*VDDIO		VDDIO+0.3	V
LOW-level input current	IIL	VIL=VSSIO	-2.5	0	+2.5	mA
HIGH-level input current	IIH	VIH=VDDIO	-2.5	0	+2.5	mA
input capacitance	CI			3.2		pF

Digital inputs with Schmitt trigger: RSTn, TM, TRIG, AUXI0…5, TESTPULSE, TESTREQX, CDROIRSTn, CKSEL, TPRSTn, TPACKX, TPACKY						
Parameter	**Symbol**	**Conditions**	**Min**	**Typ**	**Max**	**Units**
Schmitt input high voltage	VT+		0.65*VDDIO		VDDIO+0.3	V
Schmitt input low voltage	VT-		-0.3		0.35*VDDIO	V
Schmitt input hysteresis voltage	VTH			0.25		V

6.1 Input clock Characteristics

Parameter	**Symbol**	**Min**	**Typ**	**Max**	**Units**
Input clock rate (external)	Fclk			100	MHz
Input clock duty cycle (external)	dcin	45	50	55	%
Input clock RMS jitter	Jrms			1	ns
Input rise time (20% to 80% of input signal)	tri			2.5	ns
Input fall time (80% to 20% of input signal)	tfa			2.5	ns
Low-level input clock voltage (single-ended input)	V_{IL_CK}	-0.3		0.35*VDDIO	V
High-level input clock voltage (single-ended input)	V_{IH_CK}	VDDIO-0.45		VDDIO+0.3	V
Input clock capacitance*	Cin	3.8	4	4.2	pF

6.2 Parallel output interface

Parameter	Symbol	Min	Typ	Max	Units
Data rate on parallel output interface for consecutive columns (max throughput)	Fsdata_h			50	Mevents/s
Data rate on parallel output interface for consecutive rows	Fsdata_v			10	Mevents/s
Output load (assumed only capacitive)	Cout			16	pF
Output rise time (10% to 90% of signal)	tro			2.5	ns
Output fall time (90% to 10% of signal)	tfo			2.5	ns

6.3 Serial peripheral interface

Parameter	Symbol	Min	Typ	Max	Units
SPI clock frequency	Fsclk			Fclk/26	MHz
SPI clock duty cycle	Fsclk_dc		50		%
SPI pulse width low	Fsclk_pulse_lo	13*Tclk			ns
SPI pulse width high	Fsclk_pulse_hi	13*Tclk			ns
SPI input clock capacitance*	Csin	1.0	1.1	1.2	pF

7 Device Overview

7.1 Vision Sensor Core

7.1.1 Description

The vision sensor core consists of:

- Pixel array
- Readout circuitry (X and Y arbiter, and control state machine)
- Pixel biasing (bias generator)
- Pixel ROI selection circuitry
- SPI

The PPS3MVCD pixel array contains 640 (H) x 480 (V) event-driven pixels with a pixel pitch of 15um. The event-driven pixels communicate their information to the readout circuity only when a change in the scene has exceeded user-defined contrast thresholds, which allows the pixels to send out only relevant data. Consequently, temporal redundancies are removed. On the other hand, the vision sensor has an always-on full awareness feature which makes this sensor suitable to detect anything, anytime.

The pixel biasing block guarantees the event-driven pixel to work properly. Some biasing parameters can be tuned in order to set performance levels of the pixels. The modified biasing parameters are applied globally to the pixel array. For instance, power consumption, contrast threshold and pixel bandwidth are key parameters that can be tuned.

The pixel ROI selection circuitry enables the user to select multiple rectangular Regions Of Interest (ROI) in the pixel array. Pixels outside the selected ROIs do not send data. See ad-hoc section for detailed information about programming of ROIs.

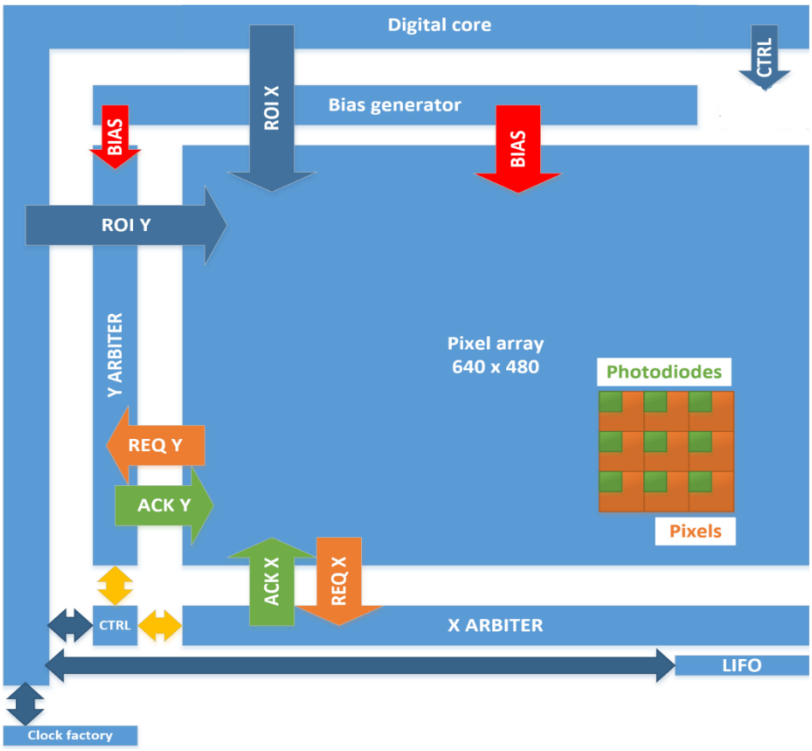


Figure 4: Device overview

7.1.2 Event-driven pixel principle behavior

The PPS3MVCD vision sensor detects temporal contrast changes of light which exceed user-defined thresholds and encodes this information into address-events that are asynchronously transmitted at the periphery via Address-Event Representation (AER). Most noteworthy features of this vision sensor are autonomous and independent pixel, low latency response, low power and data throughput dedicated to relevant data.

7.2 Address-Event Representation

Figure 5 shows a block diagram of the address event representation (AER) readout. Below illustrated is the readout sequence of a single pixel event:

1. A pixel has detected an event, meaning a change of light has exceeded either the upper or lower user-defined thresholds.
2. The pixel sends a request to the Y-dimension readout circuitry.
3. The Y readout latches the Y address.
4. The Y readout sends an acknowledge back to the pixel.
5. The pixel sends a request to the readout circuitry to the X readout.
6. The X readout latches the X address.
7. The X readout sends an acknowledge back to the pixel. The pixel is reset and is ready to detect another event.

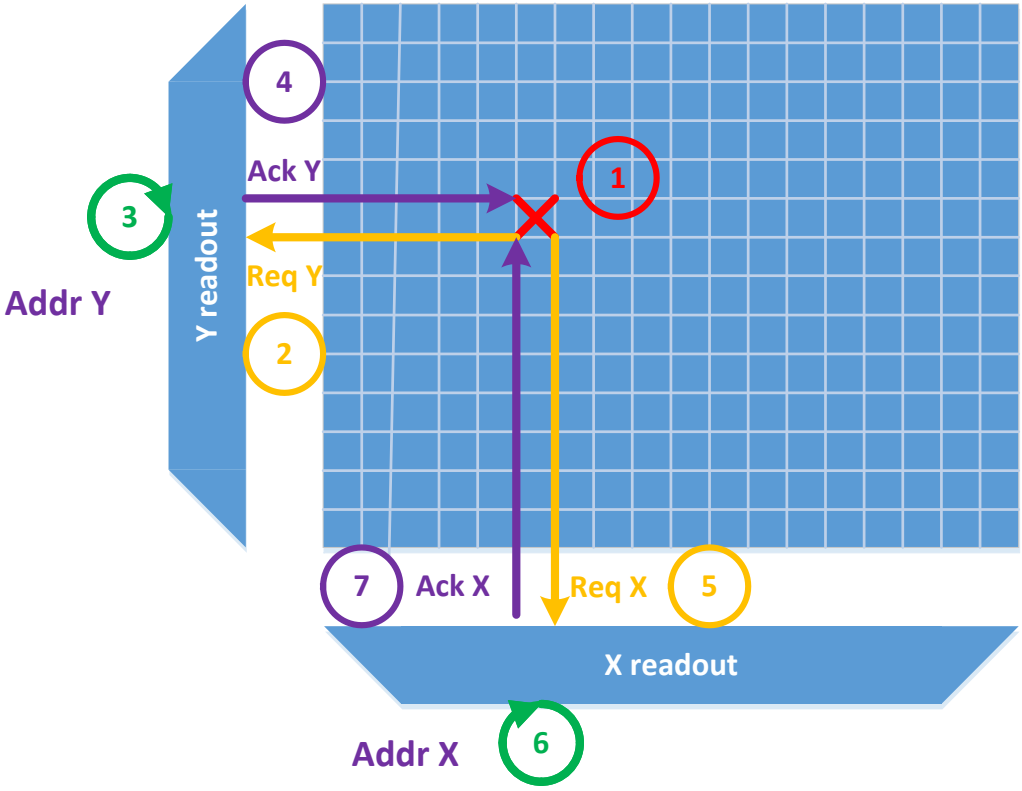


Figure 5: Address-Event Representation

7.3 Readout circuitry

The readout circuitry aim is to handle pixel requests by locking the Y and X addresses of the requests and bring them out of the vision sensor on the parallel output data bus. As for most vision sensors, the Y direction is favored compared to the X. It results that several events which occurred simultaneously within a row will be output consecutively with the same Y address. When this row is emptied, a new row can consequently be brought to the output data bus.

7.4 Input clock receiver

The single-ended 1.8-V CMOS input clock is propagated to the required blocks within the vision sensor. Maximum input clock frequency is 100 MHz. The CMOS clock signal must drive clk_in, while clk_inb must be tied to ground.

7.5 Parallel output data

The parallel output data is by default synchronous and coded on 21 bits of data + a data valid, a data ready and a clock.

7.6 Synchronous interface

To facilitate interfacing the vision sensor with some host processing, the parallel output data is synchronous. The asynchronous handshaking is done on-chip and is transparent to the end user. Consequently, the parallel output interface is composed of Y-address DY[8:0], X-address DX[9:0] and the polarity bit DPOL, in order to make distinction between ON and OFF events. The synchronous interface has a qualifying output signal called data: data_valid (DV) and an output clock signal (CLKOUT) used by the host to resynchronize the data and clk_out. Finally, the interface has an input data _ready (DR) which can hold the vision sensor output streaming. Figure 6 depicts the synchronous parallel output interface.

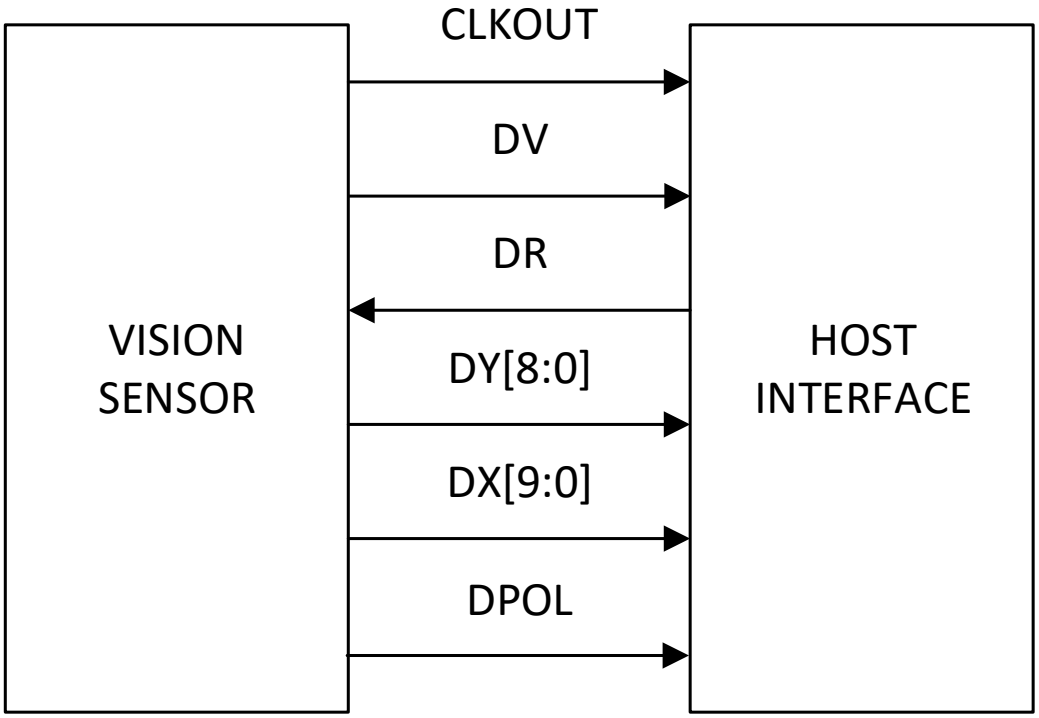


Figure 6: Synchronous parallel output data interface

When the vision sensor has a valid data or information to transmit, the data_valid signal is set high.

The following figure describes the situation in which DV is high before DR. Data is transferred when valid and ready are high on a rising edge of CLKOUT.

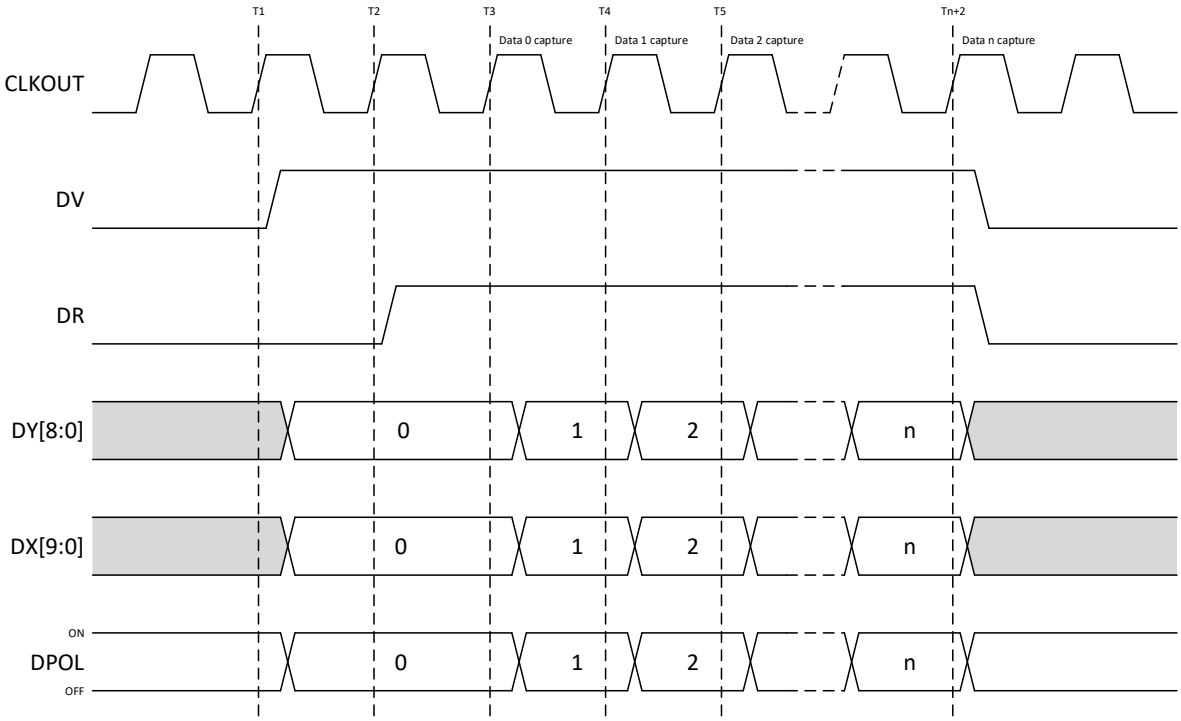


Figure 7: Synchronous parallel output interface timing diagram. DV before DR.

The following figure shows the scenario in which DR is high before DV.

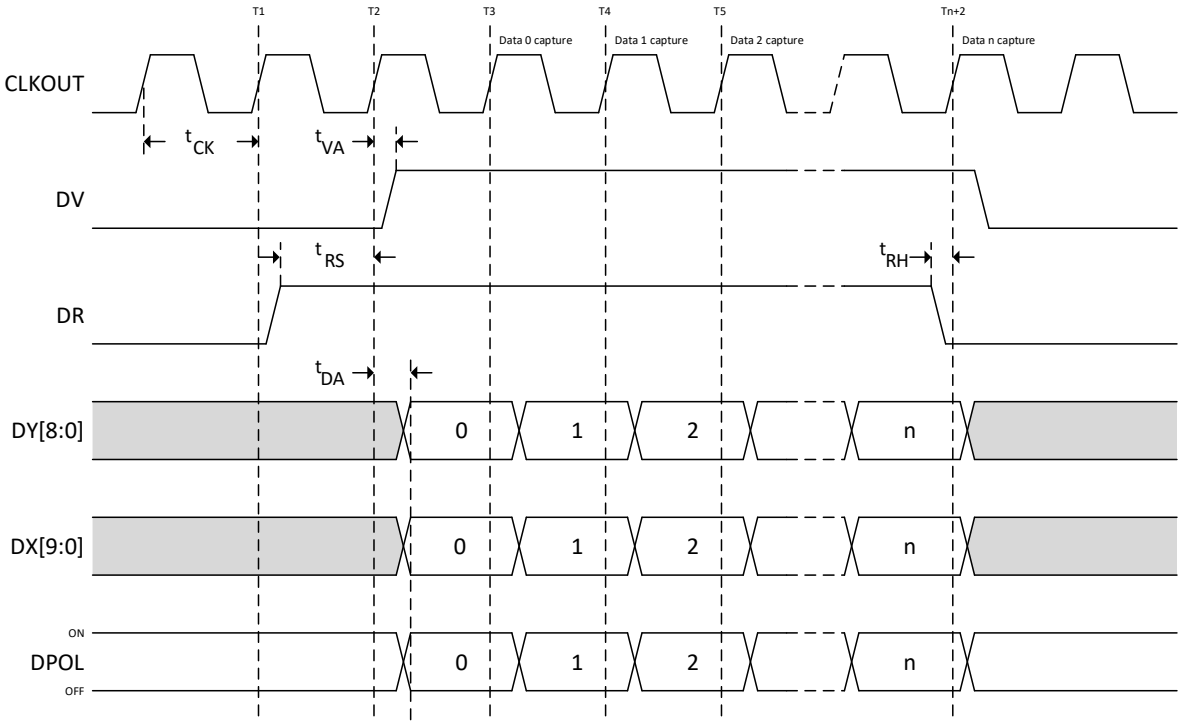


Figure 8: Synchronous parallel output interface timing diagram. DR before DV.

DPOL set high corresponds to ON events.

DPOL set low corresponds to OFF events.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
t_{VA}	DV after rising edge of CLKOUT		3.0	5.0	7.9	ns
t_{RS}	DR stable prior to rising edge of CLKOUT			3.6	6.4	ns
t_{DA}	DY[8:0], DX[9:0] and DPOL stable after rising edge of CLKOUT		2.7	5	8.1	ns
t_{RH}	DR stable after rising edge of CLKOUT		-0.16			ns
t_{CK}	Period CLKOUT		10			ns

7.7 Bias generator

The bias generator generates all required reference voltages and bias currents used on chip. Two external resistors of 1 MΩ, connected between pins rx_a and vssa and pins rx_d and vssd, are required for the bias generator to operate properly. These signals are highly sensitive and are used as reference voltages within the sensor. Routing these resistors away from digital or noisy signals is mandatory.

Moreover, having too much parasitic capacitance between these rx_a/d pads and ground can also lead to bias oscillation. It is recommended to minimize this parasitic cap to a minimum value and to target a total capacitance value below 100 pF, including parasitics.

7.8 Pixel ROI selection circuitry

The vision sensor features ROI configurations. ROI can define one or several active sub-arrays, or sub-sampling.

7.9 SPI

An SPI is embedded within the vision sensor to read/write registers which control most of the blocks.

8 Sensor operation

8.1 Power Startup Sequence

External reset RSTn must be tied low at startup to ensure the vision sensor is in defined conditions and all registers and biases are set to their default values. When RSTn is tied low, the vision sensor registers cannot be programmed, and is not functional.
Then, power supplies can ramp up. First, VDDIO which supplies the IO periphery, then VDDC which supplies the digital core, then VDDA which supplies the analog core, and finally VDDD which supplies the digital readout circuitry.
Figure 9 depicts the power startup sequence. The pads DR, CD_RSTN, EM_RSTN can be set to VDDIO at startup.

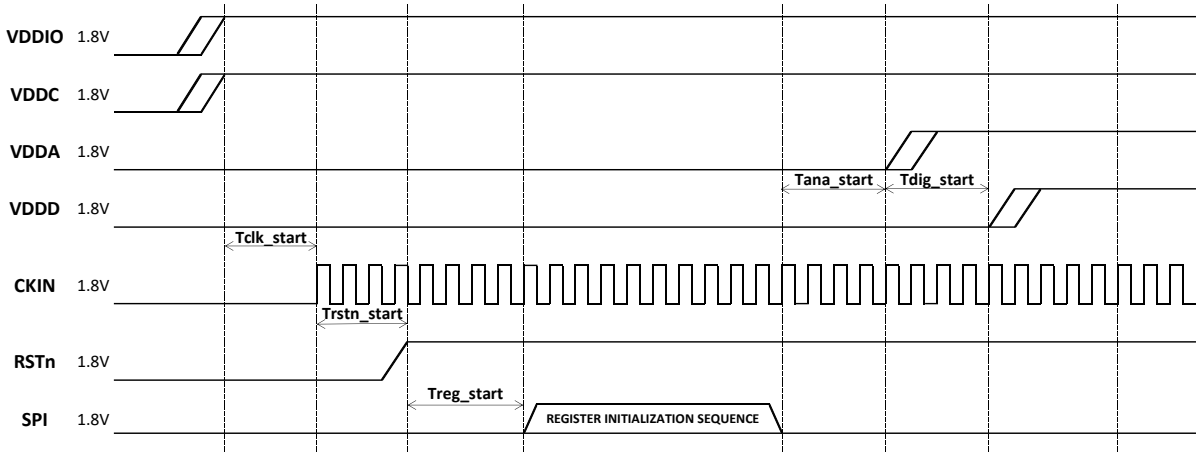


Figure 9: Power Startup Sequence.

Note: not respecting the exact sequence can lead to erratic behavior of the sensor.

Time	Minimum	Typical	Maximum
Tclk_start	1 ms	2.5 ms	10 ms*
Trstn_start	1 us	100 us	10 ms*
Treg_start	10 us	100 us	10 ms*
Tana_start	0	250 us	10 ms*
Tdig_start	0	250 us	10 ms*

*more time is no problem.

8.2 Register Initialization Sequence

During register initialization sequence, the following register data must be loaded via the SPI into the sensor.

Address	Data
0x38	0x1
0x1C	0x100
0x0	0x10
0x0	0x18
0x0	0x1a
0x40	0x140501
0x100	0x4903e1b0
0x104	0x4903e1b0
0x200	0xffffffff
0x204	0xffffffff
0x208	0xffffffff
0x20C	0xffffffff
0x210	0xffffffff
0x214	0xffffffff

0x218	0xffffffff
0x21C	0xffffffff
0x220	0xffffffff
0x224	0xffffffff
0x228	0xffffffff
0x22C	0xffffffff
0x230	0xffffffff
0x234	0xffffffff
0x238	0xffffffff
0x23C	0xffffffff
0x240	0xffffffff
0x244	0xffffffff
0x248	0xffffffff
0x24C	0xffffffff
0x250	0xffffffff
0x300	0xffffffff
0x304	0xffffffff
0x308	0xffffffff
0x30C	0xffffffff
0x310	0xffffffff
0x314	0xffffffff
0x318	0xffffffff
0x31C	0xffffffff
0x320	0xffffffff
0x324	0xffffffff
0x328	0xffffffff
0x32C	0xffffffff
0x330	0xffffffff
0x334	0xffffffff
0x338	0xffffffff
0x33C	0xffffffff
0x250	0x0
0x33C	0x0
0x4	0x2b
0x8	0x400203ff
0x44	0x400203ff
0x1C	0x10000101
0x11C	0x49014dc0
0x14C	0x7900571f
0x150	0x79125c34
0x154	0x7902ce2a
0x150	0x79125c33
0x154	0x7902ce29
0x20	0x1
0x168	0x61000000

8.3 Power Down Sequence

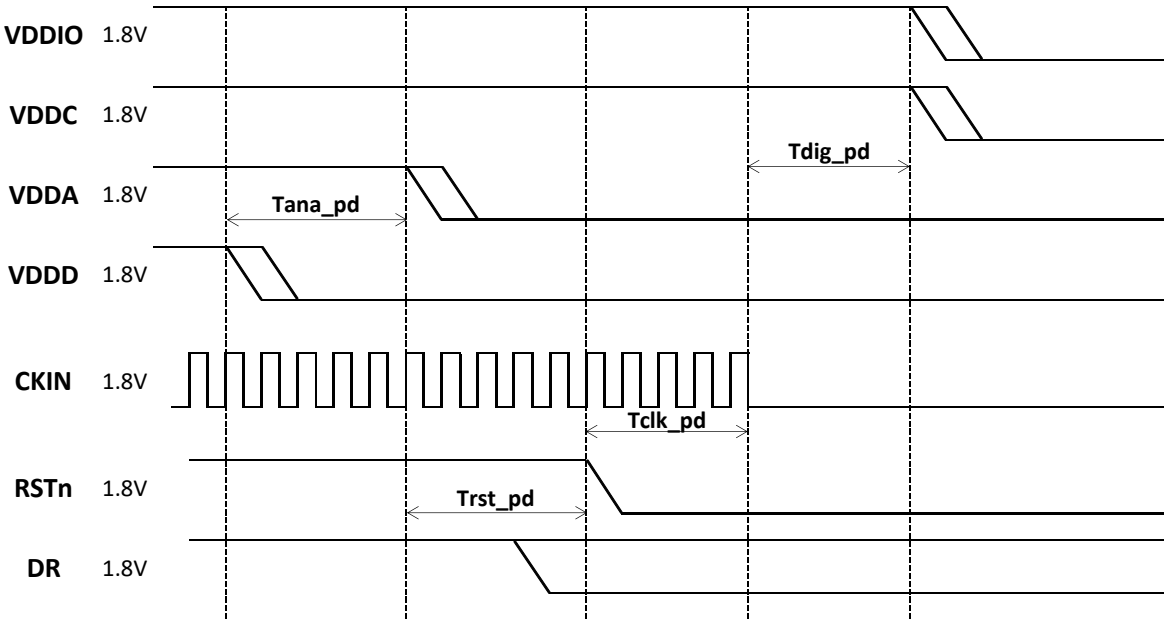


Figure 10: Power Down Sequence.

Time	Minimum	Typical	Maximum
Tana_pd	0	100 us	1 ms*
Trst_pd	1 ms	2.5 ms	10 ms*
Tclk_pd	10 us	100 us	1 ms*
Tdig_pd	0	1 ms	10 ms*

*more time is no problem.

8.4 Serial Peripheral Interface

8.4.1.1 SPI protocol description

SPI command uses a byte addressing and should be aligned on 4 bytes (if it is not the case, the address is aligned on 4 bytes automatically. Ie: a read at address 0x00000001 is going to retrieve byte 0,1,2 and 3 instead of byte 1,2,3 and 4). Therefore, an operation always takes place on the entire 32-bit register. In case of out-of-range address the error core 0xFFFFFFFC is returned and the access is dropped.

Example:

Reads at 0x0, 0x1, 0x2 or 0x3 returns register at 0x0;

Read at 0x4, 0x5, 0x6, 0x7 returns register at 0x4, and so on.

To be able to successfully communicate with the embedded SPI slave, a SPI master must respect the following specifications.

- Total data transfer size is 64 bits;
- SPI is configured in mode 3 (clock polarity CPOL = 1 and clock phase CPHA = 1). This means that MOSI line is set on the falling edge and sampled on the rising edge of SCLK. The same happens for MISO;
- A command is composed by:
    a. Preamble (8 bits): 0x40 for read, 0x00 for write;
    b. Register address (24 bits): address granularity is byte and registers are aligned on 4 bytes (therefore bits [1:0] are unused);
    c. Data (32 bits): data to be written or data read from MSB to LSB;
    d. Master shifts data on MOSI line starting from the MSB;
    e. Slave shifts data on MISO line starting from the MSB;
    f. CSn low indicates that a transfer is ongoing. Between two commands CSn must stay high for at least 4 host interface clock cycles.

Example of SPI write command:

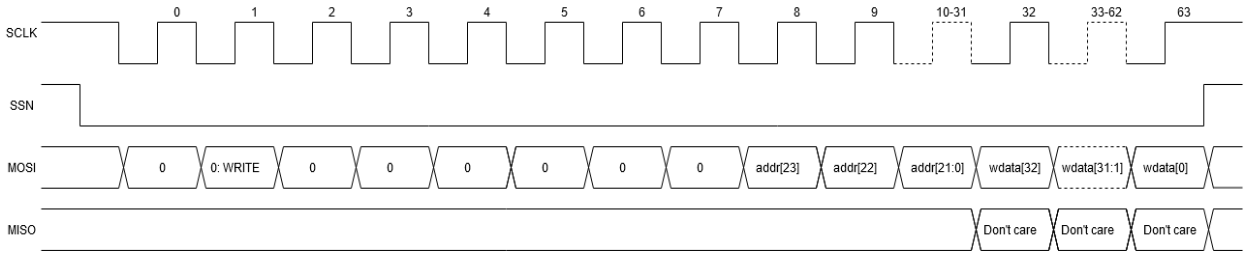


Figure 11 SPI write waveforms

Example of SPI read command:

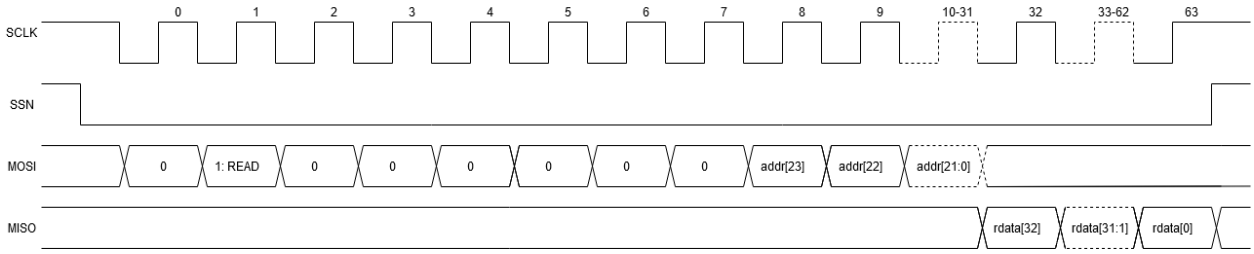


Figure 12 SPI read waveforms

Interface timing

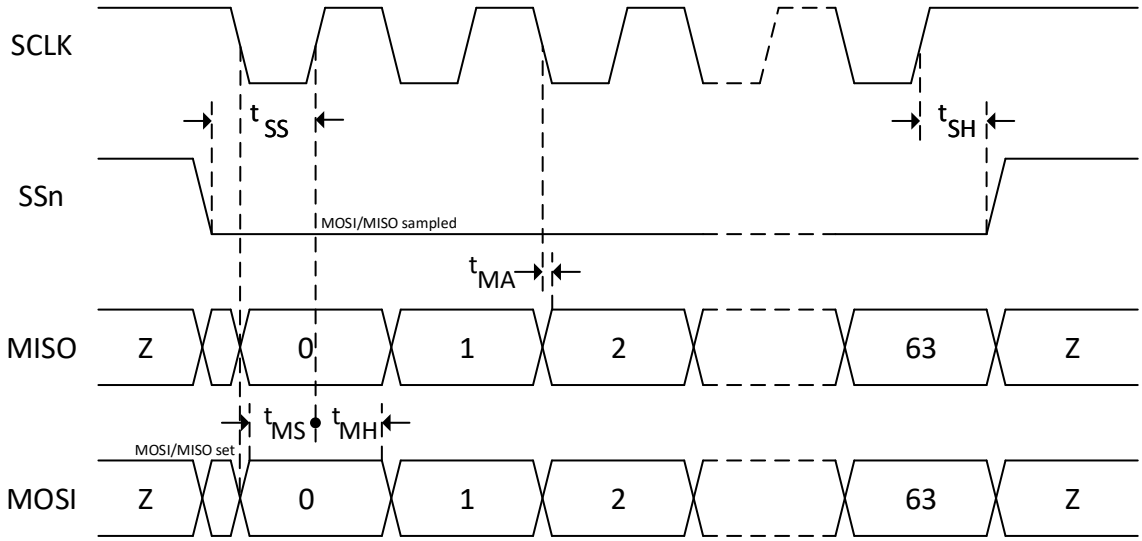


Figure 13: SPI timing diagram.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
t_{SS}	CSn setup time before SCLK rising edge		-	30	-	ns
t_{SH}	CSn hold time before SCLK rising edge		-	20	-	ns
t_{MA}	MISO line stable after SCLK falling edge		64.5	67.5	72.2	ns
t_{MS}	MOSI setup time before SCLK rising edge		-	30	-	ns
t_{MH}	MOSI hold time after SCLK rising edge		-	20	-	ns

In addition to standard SPI signals we have added a sideband wire called ERROR going from slave to master. It is routed on TPD1. It can be used within the host interface (for instance FPGA) to detect if the transfer encountered some issue. When asserted[4] it indicates:

- Address decoding error: command's address is out-of-range;
- Access permission error: trying to write to a read-only register;

Inside the slave each SPI input passes through a debouncer module used to remove signal pulses shorter than 4 clock cycles. This design choice imposes that between two consecutive SPI commands there must be at least 4 host_if_clk cycles (e.g 100MHz).

The maximum SCLK frequency is f(host_if_clk)/(13*2) where f(host_if_clk) is the clock frequency of the host interface module inside the digital core. The main digital core clock has a maximum frequency of 100MHz.

[4] Asserted is logic high hence signal at VDD.

8.4.2 SPI registers

The host can access the full sensor's register bank is given in the Table below.

Table 1: Register bank.

Register bank name	Register name	Register address	Default value
main control	*Reserved*	0x000	0x00000000
	roi_ctrl	0x004	0x0000000B
	Reserved	0x008	0x40000BFF
	Reserved	0x00C	0x00000040
	Reserved	0x010	0x00000006
	lifo_ctrl	0x014	0x00000000
	Reserved	0x018	0x90200C02
	spare_ctrl	0x01C	0x00000000
	Reserved	0x020	0x00000000
	Reserved	0x024	0x00000000
	Reserved	0x028	0x00000000
	Reserved	0x02C	0x00000000
	Reserved	0x030	0x00000000
	Reserved	0x034	0x00000001
	clk_out_ctrl	0x038	0x00000000
	stst_ctrl	0x03C	0x00000000
	Reserved	0x040	0x00140500
	Reserved	0x044	0x400003FF
	nfl_ctrl	0x048	0x000000C8
	nfl_threshold	0x04C	0x00190032
	nfl_cnt	0x050	0x00000000
bias vector control	*Reserved*	0x100	0x4903E17F
	Reserved	0x104	0x4903E17F
	Reserved	0x108	0x4900329B
	Reserved	0x10C	0x49001BAA
	Reserved	0x110	0x49014D8D
	Reserved	0x114	0x690DB771
	Reserved	0x118	0x49014D8D
	Reserved	0x11C	0x49014D8D
	Reserved	0x120	0x69002C3F
	Reserved	0x124	0x69005437
	Reserved	0x128	0x59166CB8
	Reserved	0x12C	0x79000000
	Reserved	0x130	0x590000F8
	Reserved	0x134	0x7102CE2A
	Reserved	0x138	0x7108002F
	Reserved	0x13C	0x71008321
	Reserved	0x140	0x590004E3
	Reserved	0x144	0x590000F1
	Reserved	0x148	0x49014D8D
	bias_diff_off_ctrl	0x14C	0x7100571F
	bias_diff_on_ctrl	0x150	0x71125C34
	bias_diff_ctrl	0x154	0x7102CE2A
	bias_fo_ctrl	0x158	0x610036D6
	bias_pr_ctrl	0x15C	0x510345B1
	bias_refr_ctrl	0x160	0x51004FD4
	bias_hpf	0x164	0x510002D4
	Reserved	0x168	0x61013254
cd roi x vector control	cd_roi_x00	0x200	0xFFFFFFFF
	cd_roi_x01	0x204	0xFFFFFFFF
	cd_roi_x02	0x208	0xFFFFFFFF
	cd_roi_x03	0x20C	0xFFFFFFFF
	cd_roi_x04	0x210	0xFFFFFFFF
	cd_roi_x05	0x214	0xFFFFFFFF
	cd_roi_x06	0x218	0xFFFFFFFF
	cd_roi_x07	0x21C	0xFFFFFFFF

	cd_roi_x08	0x220	0xFFFFFFFF
	cd_roi_x09	0x224	0xFFFFFFFF
	cd_roi_x10	0x228	0xFFFFFFFF
	cd_roi_x11	0x22C	0xFFFFFFFF
	cd_roi_x12	0x230	0xFFFFFFFF
	cd_roi_x13	0x234	0xFFFFFFFF
	cd_roi_x14	0x238	0xFFFFFFFF
	cd_roi_x15	0x23C	0xFFFFFFFF
	cd_roi_x16	0x240	0xFFFFFFFF
	cd_roi_x17	0x244	0xFFFFFFFF
	cd_roi_x18	0x248	0xFFFFFFFF
	cd_roi_x19	0x24C	0xFFFFFFFF
	Reserved	0x250	0x0000000
cd roi y vector control	cd_roi_y00	0x300	0xFFFFFFFF
	cd_roi_y01	0x304	0xFFFFFFFF
	cd_roi_y02	0x308	0xFFFFFFFF
	cd_roi_y03	0x30C	0xFFFFFFFF
	cd_roi_y04	0x310	0xFFFFFFFF
	cd_roi_y05	0x314	0xFFFFFFFF
	cd_roi_y06	0x318	0xFFFFFFFF
	cd_roi_y07	0x31C	0xFFFFFFFF
	cd_roi_y08	0x320	0xFFFFFFFF
	cd_roi_y09	0x324	0xFFFFFFFF
	cd_roi_y10	0x328	0xFFFFFFFF
	cd_roi_y11	0x32C	0xFFFFFFFF
	cd_roi_y12	0x330	0xFFFFFFFF
	cd_roi_y13	0x334	0xFFFFFFFF
	cd_roi_y14	0x338	0xFFFFFFFF
	Reserved	0x33C	0x00000000
Reserved vector control	*Reserved*	0x400	0xFFFFFFFF
	Reserved	0x404	0xFFFFFFFF
	Reserved	0x408	0xFFFFFFFF
	Reserved	0x40C	0xFFFFFFFF
	Reserved	0x410	0xFFFFFFFF
	Reserved	0x414	0xFFFFFFFF
	Reserved	0x418	0xFFFFFFFF
	Reserved	0x41C	0xFFFFFFFF
	Reserved	0x420	0xFFFFFFFF
	Reserved	0x424	0xFFFFFFFF
	Reserved	0x428	0xFFFFFFFF
	Reserved	0x42C	0xFFFFFFFF
	Reserved	0x430	0xFFFFFFFF
	Reserved	0x434	0xFFFFFFFF
	Reserved	0x438	0xFFFFFFFF
	Reserved	0x43C	0xFFFFFFFF
	Reserved	0x440	0xFFFFFFFF
	Reserved	0x444	0xFFFFFFFF
	Reserved	0x448	0xFFFFFFFF
	Reserved	0x44C	0xFFFFFFFF
	Reserved	0x450	0x00000000
Reserved vector control	*Reserved*	0x500	0xFFFFFFFF
	Reserved	0x504	0xFFFFFFFF
	Reserved	0x508	0xFFFFFFFF
	Reserved	0x50C	0xFFFFFFFF
	Reserved	0x510	0xFFFFFFFF
	Reserved	0x514	0xFFFFFFFF
	Reserved	0x518	0xFFFFFFFF
	Reserved	0x51C	0xFFFFFFFF
	Reserved	0x520	0xFFFFFFFF

	Reserved	0x524	0xFFFFFFFF
	Reserved	0x528	0xFFFFFFFF
	Reserved	0x52C	0xFFFFFFFF
	Reserved	0x530	0xFFFFFFFF
	Reserved	0x534	0xFFFFFFFF
	Reserved	0x538	0xFFFFFFFF
	Reserved	0x53C	0x00000000

The register map ends at the address 0x53F.

All reserved registers should be left untouched except instructed differently in this datasheet. For instance, during initialization sequence, some of these reserved registers are overwritten.

PROPHESEE cannot guarantee sensor behavior if these register values are updated.

8.4.3 Output clock control

Address		Register Name	ReadMask	WriteMask	ResetValue	Comment
0x00000038		clk_out_ctrl	0x000007FF	0x000007FF	0x00000000	
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**Comment**
	0: 0	clk_out_en	01	RW	0x00000000	enable clk on the output parallel interface
	1: 1	clk_gate_bypass	01	RW	0x00000000	bypass the output clk gate. clk out is always on
	2: 2	clk_out_pol	01	RW	0x00000000	invert clk out polarity
	10: 3	clk_timeout	08	RW	0x00000000	maintain the clock for a programmable number of cycles after the activity is over

The output clock of the parallel interface is controlled directly by the digital core.

Output clock is disabled by default and shall be enabled to transmit data (clk_out_en register). If clock gating is bypassed (clk_gate_bypass=1), the output clock is always-on regardless of the activity of the valid signal as described in the following figure.

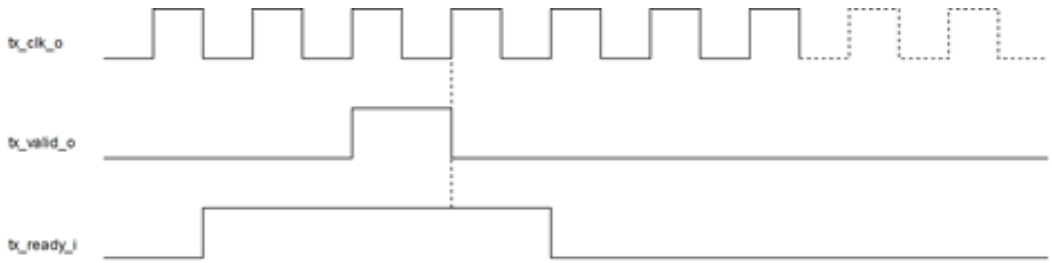


Figure 14 Always on output clock

When enabled and not bypassed (clk_gate_bypass=0) the output clock is switched on at least 2 cycles before the valid transfer edge. Also notice that clk_timeout=0, therefore the clock is switched off right after the transfer cycle. The following figure describes the above-mentioned configuration.

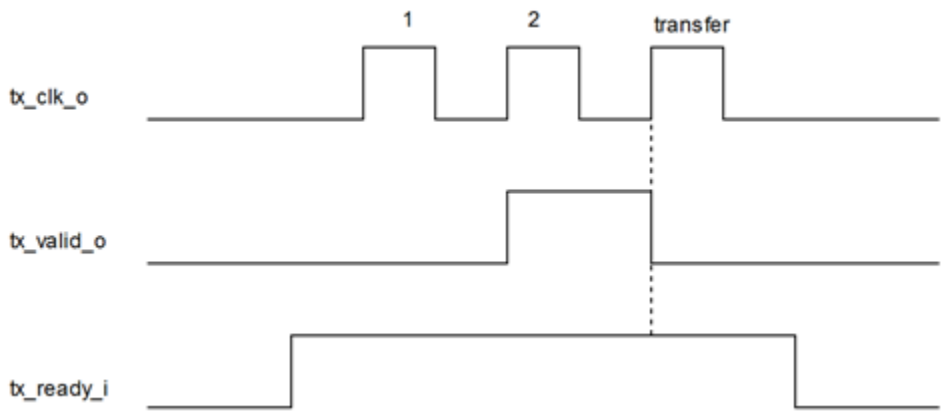


Figure 15 Shortest clock pulse possible

Clk_timeout register indicates how many cycles the output clock shall be kept on after a data transfer. This feature is meant to help data synchronization on the RX side. In the following example clk_timeout register is set to decimal 4.

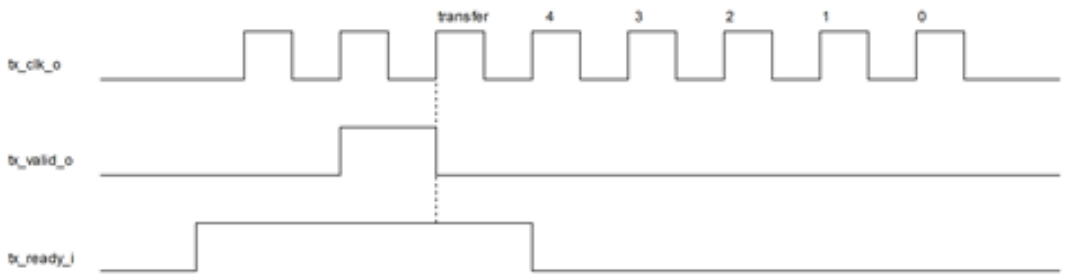


Figure 16 Output clock with clk_timeout

Polarity of the output clock can be inverter thanks to clk_out_pol register.

8.4.4 Self-test control

Address		Register Name	ReadMask	WriteMask	ResetValue	Comment
0x0000003C		stst_ctrl	0x00000FFF	0x00000FFF	0x00000000	
	high:low	Field Name	size	Attribute	ResetValue	Comment
	1:0	stst_en	02	RW	0x00000000	enable the self-test module in 3 modes: 00 disable, 01 lfsr, 10 incr, 11 incr with random valid
	11:2	stst_div_factor	10	RW	0x00000000	event rate divider

Data transmission on the parallel output interface can be debugged by activating a pattern generator and checking that the expected sequences are correctly received on the host side. Three patterns are supported:

- Pseudo random based on 21 bit galois lfsr. Taps: 0x1C8000. Start value 0x000001. Programmable rate from 100Mevt/s to 100Kevt/s;
- incremental counter: binary counter doing a +1 at every cycle. Programmable rate from 100Mevt/s to 100Kevt/s;
- incremental counter with random valid: binary counter with randomly toggling valid signal;

A complete round of the counter lasts 2^{21} cycles if configured in mode incremental and (2^{21})-1 if configured in mode lfsr.
Between two consecutive iterations we insert a pause of 2048 cycles = 20.480us to ease the synchronization.
Enabling the self-pattern block with “stst_en” = 1, disactivates the event data pipe. And conversely, disabling the self-pattern block with “stst_en” = 0, enables back the event data pipe.
The input ready is taken into account in the logic except for the rate counter and during the pause between two iterations. They keep running even if the ready is low.
If the input ready toggles the spacing between valid data in input cannot be guarantee because the rate counter are not conditioned by the ready signal.
To switch between modes is mandatory to disable the counter first and then select the new mode.
The speed of the pattern generator can be controlled via the stst_div_factor register. A valid data is produced every “div_factor” + 1 cycle. The stst_div_factor can be modified dynamically while the pattern generator is enabled. The following code snippet show the lfsr and incr pseudo-code.

```
#LFSR PSEUDO CODE
  taps=0x001C8000
  lfsr=0x00000001
  D[20:0] = 0
 for i in range(0, 2**21, 1):
    lfsr_bit0 = lfsr & 0x00000001
    if lfsr_bit0 == 0x00000001:
        lfsr_tmp = (2**21)-1
    else:
        lfsr_tmp = 0

    lfsr_tmp = lfsr_tmp and taps
    lfsr = xor(lfsr>>1, lfsr_tmp)
    lfsr = lfsr & 0x001FFFFF
    D = lfsr


# INCR PSEUDO CODE
  D[20:0] = 0
  for i in range(0, 2**21, 1):
    D[20:0] = D[20:0] + 1
```

The 21 bits from the counter are mapped as follow on the CMOS parallel interface:

Counter bits	CMOS parallel interface output pin
D[20]	DPOL
D[19]	DTY
D[18:10]	DY8:DY0
D[9:0]	DX9:DX0

8.4.5 On-the-fly sensor configuration

Configuration of the sensor consists of programming registers, bias generators and ROI. All biases are independent from each other. As ROIs have shadow registers, a complete set of ROIs can be programmed via SPI and then transferred at once via a single SPI command. This feature makes on-the-fly programming possible. Values being effective into the sensor once correctly programmed.

8.5 Region Of Interest control

Address		Register Name	ReadMask	WriteMask	ResetValue	Comment
0x00000004		roi_ctrl	0x000001BB	0x000001BB	0x0000000B	
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**Comment**
	1:1	roi_cd_en	01	RW	0x00000001	Enable ROI CD 0 : CD full array 1 : CD ROI
	3:3	roi_cd_rstn	01	RW	0x00000001	0: Resets the whole CD ROI X and Y arrays
	4:4	*Reserved*	01	RW	0x00000000	
	5:5	roi_cd_shadow_trigger	01	RW	0x00000000	Latch the configured ROI CD into the shadow register
	7:7	*Reserved*	01	RW	0x00000000	
	8:8	roi_cd_scan_en	01	RW	0x00000000	Any modification applied to roi cd register is directly forwarded to the corresponding shadow register

It is possible to define one or several Region of Interests (ROI) on the vision sensor. The ROI is programmed thanks to cd roi x vector control / cd roi y vector control. The ROI pattern consists of 1120 bits of which 640 bits are used for X-address selection and 480 bits for Y-address selection. The ROI also has a shadow register so that a 1120-bit pattern can be written and loaded only when the programming sequence is over. This shadow register enables update of single bits or generation of dynamic sequences on any bits.

Table 2: ROI CD Karnaugh map

roi_cd_rstn	roi_cd_en	roi_cd_x/y bit register	CD pixel status
0	0	0	0 = Reset
0	0	1	0 = Reset
0	1	0	0 = Reset
0	1	1	0 = Reset
1	0	0	1 = Active
1	0	1	1 = Active
1	1	0	0 = Reset
1	1	1	1 = Active

Equation of Table 2 is: $cd_pixel_status = roi_cd_rstn \cdot \overline{roi_cd_en} + roi_cd_rstn \cdot roi_cd_reg$

To reset all CD pixels, roi_cd_rstn must be set low.

roi_cd_rstn can be provided either by an external pad or an SPI register.

In order to have all CD pixels functional, signal roi_cd_rstb must be set high and signal roi_cd_en must be set low. Hence, register values do not influence CD part of pixels.

Figure 17 shows an example of CD ROI array programming. For this example, CD ROI is enabled thanks to roi_cd_rstb and roi_cd_en both high. A CD pixel will be in reset as soon as either is corresponding X register or Y register is set low. Otherwise, the CD pixel is active and behaves as normal.

Y registers								
0	Reset	Reset	Reset	Reset	Reset	Reset	Reset	Reset
0	Reset	Reset	Reset	Reset	Reset	Reset	Reset	Reset
1	Reset	Active	Active	Active	Active	Active	Active	Reset
1	Reset	Active	Active	Active	Active	Active	Active	Reset
1	Reset	Active	Active	Active	Active	Active	Active	Reset
1	Reset	Active	Active	Active	Active	Active	Active	Reset
0	Reset	Reset	Reset	Reset	Reset	Reset	Reset	Reset
0	Reset	Reset	Reset	Reset	Reset	Reset	Reset	Reset
X registers	0	1	1	1	1	1	1	0

Figure 17: Example of CD ROI array programming

8.5.1 Multiple Region Of Interest

Multiple Region Of Interests can be chosen but some combinations imply some side effects.

Figure 18 shows an example of multiple ROIs. For instance, defining one ROI located in top left corner and one ROI located in bottom right corner (depicted in blue) implies 2 other ROIs located in top right and bottom left corners (shown in orange). This combination of multiple ROIs can easily be filtered out in post-processing.

Y registers \ X registers								
0	Reset	Reset	Reset	Reset	Reset	Reset	Reset	Reset
1	Reset	Active	Active	Reset	Reset	Active	Active	Reset
1	Reset	Active	Active	Reset	Reset	Active	Active	Reset
0	Reset	Reset	Reset	Reset	Reset	Reset	Reset	Reset
0	Reset	Reset	Reset	Reset	Reset	Reset	Reset	Reset
1	Reset	Active	Active	Reset	Reset	Active	Active	Reset
1	Reset	Active	Active	Reset	Reset	Active	Active	Reset
0	Reset	Reset	Reset	Reset	Reset	Reset	Reset	Reset
	0	1	1	0	0	1	1	0

Figure 18: Example of multiple ROIs

8.5.2 Sub-sampling

Thanks to the ROI, any kind of subsampling within the VGA resolution is possible. Subsampling does not impact static power consumption. However, dynamic power consumption is likely to be reduced due to a drop of activity.

8.5.3 ROI X control

Address		Register Name	ReadMask	WriteMask	ResetValue	Comment
0x00000200		cd_roi_x00	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000204		cd_roi_x01	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000208		cd_roi_x02	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x0000020C		cd_roi_x03	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000210		cd_roi_x04	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000214		cd_roi_x05	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000218		cd_roi_x06	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x0000021C		cd_roi_x07	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000220		cd_roi_x08	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000224		cd_roi_x09	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	

0x00000228		cd_roi_x10	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x0000022C		cd_roi_x11	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000230		cd_roi_x12	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000234		cd_roi_x13	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000238		cd_roi_x14	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x0000023C		cd_roi_x15	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000240		cd_roi_x16	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000244		cd_roi_x17	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000248		cd_roi_x18	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x0000024C		cd_roi_x19	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**Comment**
	31:0	val	32	RW	0xFFFFFFFF	CD ROI X | 0: disable | 1: enable | Requires CD shadow trigger to get effective

8.5.4 ROI Y control

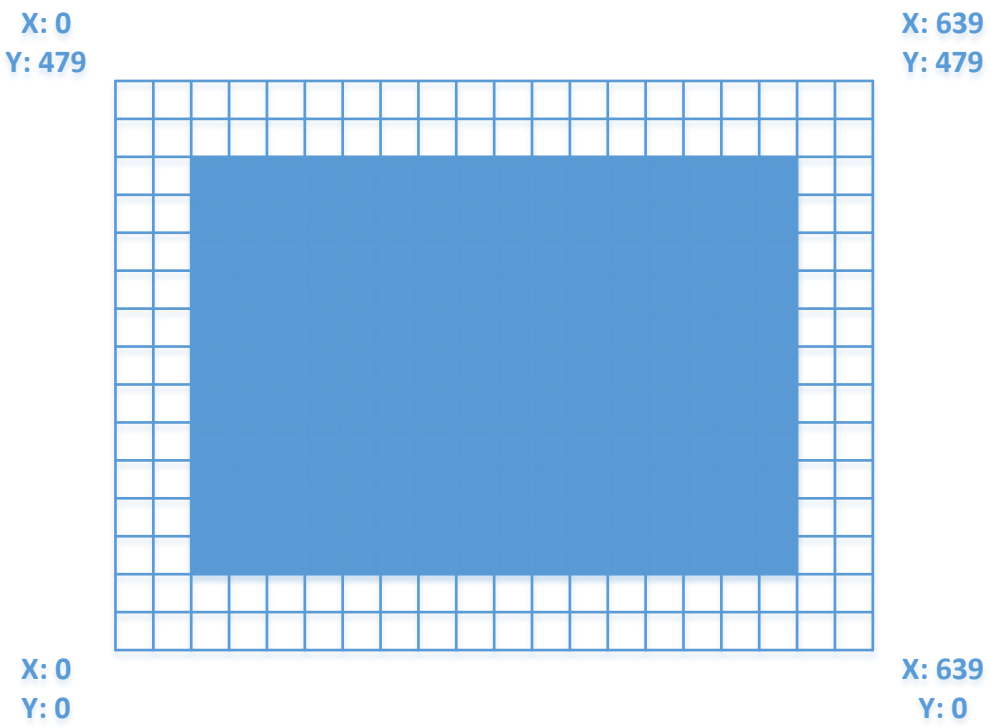
Address		Register Name	ReadMask	WriteMask	ResetValue	Comment
0x00000300		cd_roi_y00	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000304		cd_roi_y01	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000308		cd_roi_y02	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x0000030C		cd_roi_y03	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000310		cd_roi_y04	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000314		cd_roi_y05	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000318		cd_roi_y06	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x0000031C		cd_roi_y07	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000320		cd_roi_y08	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000324		cd_roi_y09	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000328		cd_roi_y10	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x0000032C		cd_roi_y11	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000330		cd_roi_y12	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000334		cd_roi_y13	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
0x00000338		cd_roi_y14	0xFFFFFFFF	0xFFFFFFFF	0xFFFFFFFF	
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**Comment**
	31:0	val	32	RW	0xFFFFFFFF	CD ROI Y | 0: disable | 1: enable | Requires CD shadow trigger to get effective

8.5.5 Example of a square ROI programming

Once sensor has been initialized.

Register address	Register value	Waiting time between consecutive commands [us]	Comments
0x200	0x00000000	0	Program roi_cd_x_vect to disable 32 columns from x=0 to 31
0x204	0x00000000	0	Program roi_cd_x_vect to disable 32 columns from x=32 to 63
0x208	0x00000000	0	Program roi_cd_x_vect to activate 32 columns from x=576 to 607
0x20C	0x00000000	0	Program roi_cd_x_vect to activate 32 columns from x=608 to 639
0x300	0x00000000	0	Program roi_cd_y_vect to activate 32 rows from y=0 to 31
0x304	0x00000000	0	Program roi_cd_y_vect to activate 32 rows from y=32 to 63
0x334	0x00000000	0	Program roi_cd_y_vect to activate 32 rows from y=416 to 447
0x338	0x00000000	0	Program roi_cd_y_vect to activate 32 rows from y=448 to 479
0x004	0x0000002B	0	Enable CD roi and trigger transfer values to the shadow registers

The SPI commands described above will give the following 512x352 pixel ROI pattern:



8.6 Event filter control

Address		Register Name	ReadMask	WriteMask	ResetValue	Comment
0x00000048		nfl_ctrl	0x000000FF	0x000000FF	0x000000C8	
	high:low	Field Name	size	Attribute	ResetValue	Comment
	0:0	nfl_en	01	RW	0x00000000	noise filter enable
	7:1	pre_cnt_thresh	07	RW	0x00000064	number of clock cycles needed to create the 1 micro second tick

This register provides control over sensor's "event filter" or "noise filter" feature. The purpose of this functionality is to transmit events only if beyond a certain threshold to allow to save as much as power as possible. The functionality is disabled by default. The user can enable it by setting nfl_en bit.

The noise filter calculates the event rate using a discrete time-shifting window. The filter counts the number of events during a programmable time (called "period"). If the counted events are above the programmed threshold the sensor output is enabled. On the other hand, if the counted events are below the programmed threshold, sensor's output is disabled until the event threshold is reached again.

8.6.1 Event filter threshold control

Address		Register Name	ReadMask	WriteMask	ResetValue	Comment
0x0000004C		nfl_thresh	0x00FFFFFF	0x00FFFFFF	0x00190032	
	high:low	Field Name	size	Attribute	ResetValue	Comment
	13:0	evt_thresh	14	RW	0x00000032	event per period threshold above which the output is enabled

	23:14	period_cnt_thresh	10	RW	0x00000064	period counter threshold in microseconds

Number of events per event period is programmable via the register evt_thresh.

Period length in 'us' is programmable in the period_cnt_thresh register. Available[5] values range from 1 to 1023. However, the valid[6] range is from 100 to 1023. The microsecond tick can be adjusted using the "pre_cnt_thresh" (i.e. in case the sensor runs at frequency lower than 100MHz). User has to program the event threshold (measured in number of event per period) below which events are not transmitted. The available event rate ranges from 25kevt/s to 10Mevt/s.

The user is responsible for programming in a consistent way the above-mentioned parameters in order to reach the best tradeoff between latency and precision. By default, the block is configured to output events above 500kevt/s (50 events every 100us).

8.6.2 Event counter control

Address		Register Name	ReadMask	WriteMask	ResetValue	Comment
0x00000050		nfl_cnt	0x00003FFF	0x00000000	0x00000000	
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**Comment**
	13:0	evt_thresh	14	RV	0x00000000	events counted during the last period

The number of events generated by the sensor at the output during a window "period_cnt_thresh" is dumped into the register evt_thresh for monitoring and debug purposes.

The longer the "period_cnt_thresh" window is, the more accurate the counting but the lower the refresh rate.

8.7 Global illuminance

The global luminance feature, also called LIFO which acronym stands for Light Input Frequency Output, is an on-chip circuit that converts the instantaneous photocurrent of the whole pixel array into a pulse width modulated signal that can be read out from the SPI bits "lifo_counter<25:0>" or via the external pin FOUT (J13). The signal encodes the absolute illumination of the scene in the field of view of the sensor. The information can be used to control camera system parameters or sensor chip settings.

The illumination measurement is controlled via the SPI.

Address		Register Name	ReadMask	WriteMask	ResetValue	Comment
0x00000014		lifo_ctrl	0xE3FFFFFF	0xC0000000	0x00000000	
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**Comment**
	25:0	lifo_counter	26	RV	0x00000000	Pulsewidth measured value: Max illuminance measurement pulse = 5*E-01s, min Tclk = 1*E-08 gives a max counter of 5*E07

[5] Available range corresponds to the complete range provided. It corresponds to the number of register bits that are present in the sensor to program a value.
[6] Valid range is defined as the range for which the sensor produces valid and expected output data.

	29:29	lifo_counter_valid	01	RV	0x00000000	Pulsewidth measured value is valid
	30:30	lifo_cnt_en	01	RW	0x00000000	Enable pulsewidth counter
	31:31	lifo_en	01	RW	0x00000000	Enable illuminance measurement 0: photocurrent measurement inactive 1: photocurrent measurement active

The register "lifo_en" controls the analog part of the LIFO block. When this signal is set to '0', photocurrent measurement is disabled and the external pin Fout is clamped to 1.8V. When "lifo_en" is set to '1', photocurrent measurement is enabled and the external pin Fout outputs a pulse width modulated signal which high-level pulse duration is proportional to total illumination of the chip as depicted on Figure 19. Typical duration of the low-level pulse is set to 500 ns.

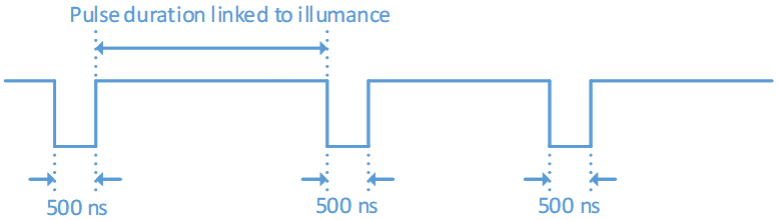


Figure 19: Typical global illuminance Fout signal waveform.

For the LIFO to operate properly, it is mandatory to leave the external pin IPHOTO (K13) floating.

Each time, a LIFO count must be acquired, the register lifo_cnt_en must be set to 0 and then to 1 to trigger a new counter acquisition. A new counter acquisition is confirmed by lifo_counter valid equals to 1.

depicts the LIFO count function of a 505nm light applied to the whole matrix. The whole photocurrent of the array is contributing to the LIFO count. The LIFO counter has a 26-bit depth which gives a maximum count of 6 7108 864. One count is equal to 10ns. This maximum count is reached around 10 uW/m2 and it explains why the LIFO clamps for lowest light level. A power fit is provided in the figure to extrapolate the internal behavior. The counter can overflow during the counting as the counter from Fout rising up to Fout falling edge. llifo_counter_valid flips to 1 when Fout falling edge is detected.

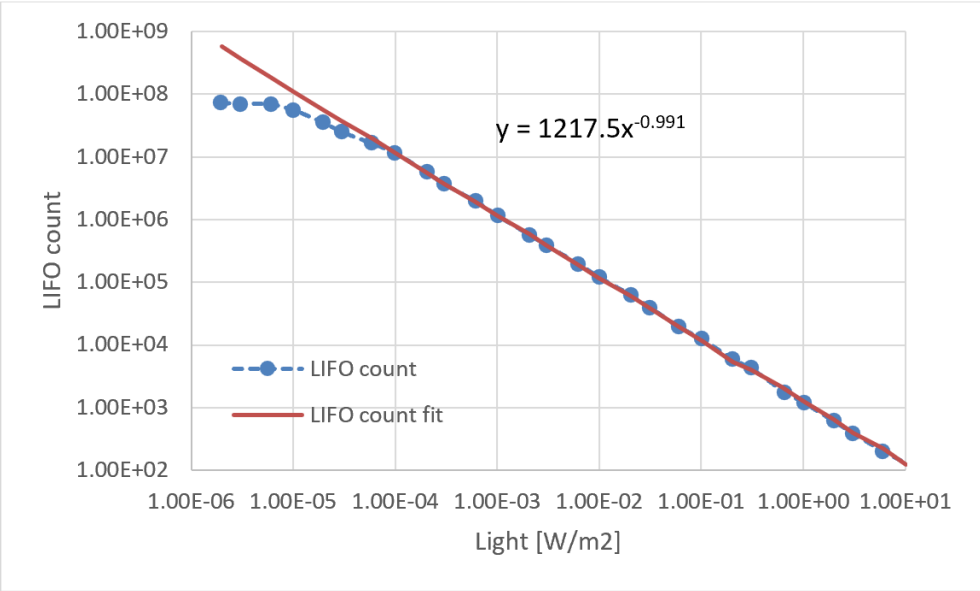


Figure 21: LIFO count function of a 505nm light.

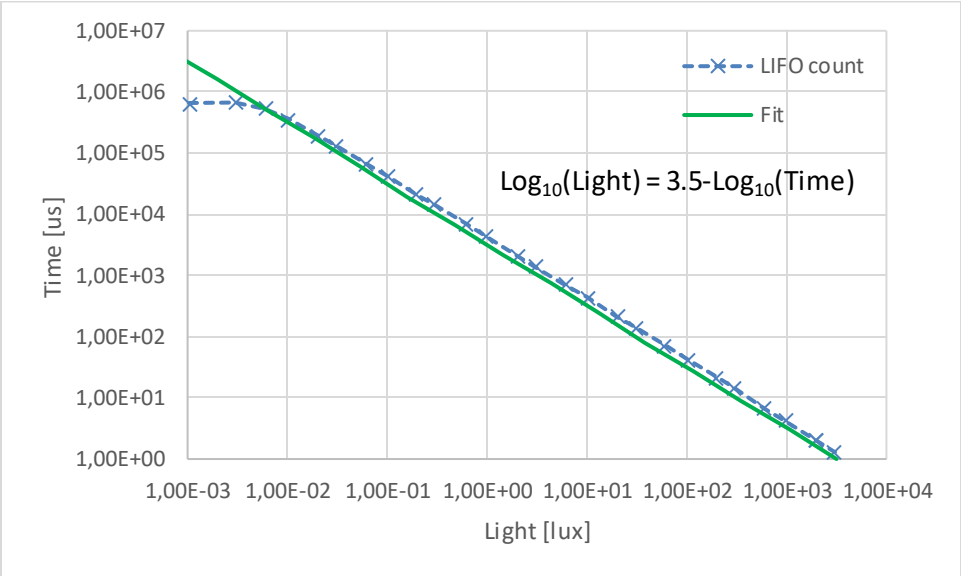


Figure 22: Time function of light.

8.8 Figure 23: Time function of light.Quantum Efficiency (QE)

It is possible to derive the Quantum Efficiency (QE) from the LIFO data. By performing the LIFO measurement with different emitting diodes and knowing the LED emission spectra and the light level given by the fitting equation, we can calculate the QE. The result is shown in Table 3. Each point in the table represents a different LED source.

	455nm	505nm	625nm	850nm	940nm
QE from LIFO	37.6%	39.1%	37.3%	14.8%	5.3%

Table 3 Quantum Efficiency measured from LIFO for 5 different LED sources.

8.9 Bias vector control

This section describes the "user" biases as shown in Figure 24.

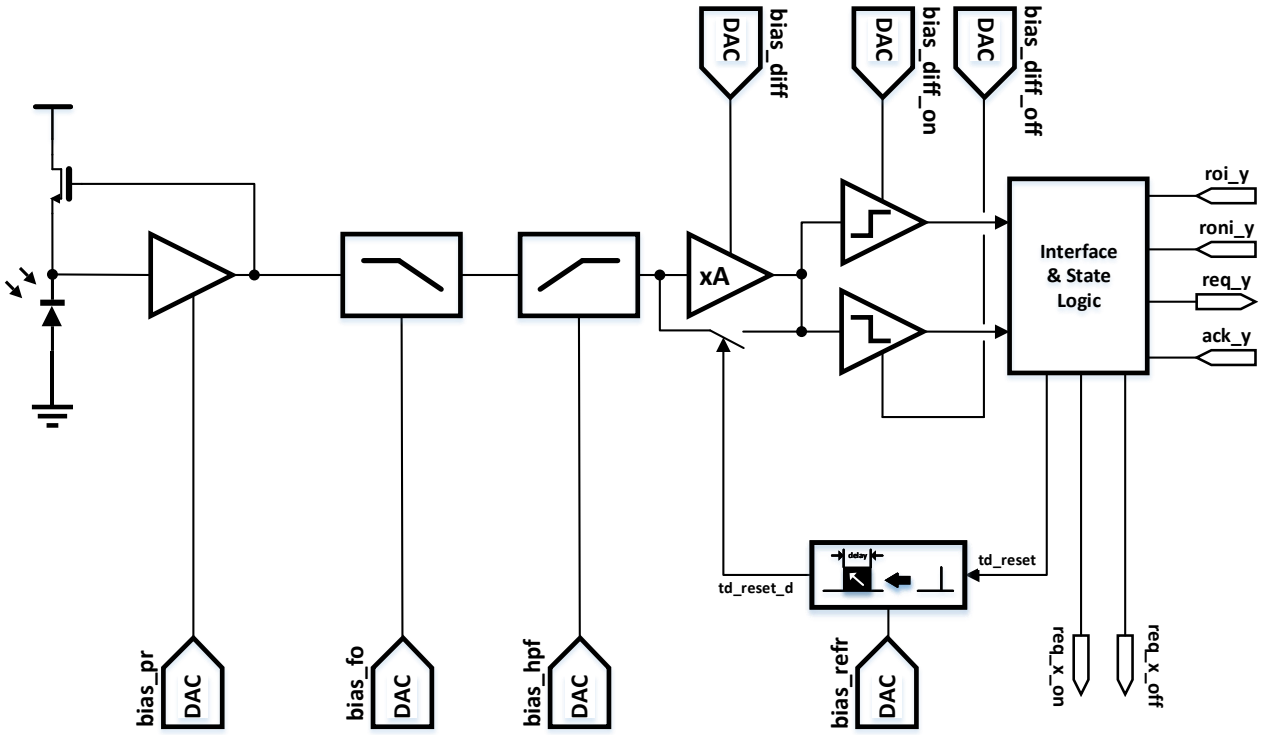


Figure 24: Event-based pixel block diagram.

For more details or support about this section, please contact PROPHESEE and consult PROPHESEE pixel biasing application note.

8.9.1 Bias pr control

This bias controls the front-end part of the pixel, the photoreceptor.

Address		Register Name	ReadMask	WriteMask	ResetValue	FunctionalValue	Comment
0x0000015C		bias_pr_ctrl	0xFFFFFFFF	0xFFFFFFFF	0x510345B1	0x510345B1	Photo-receptor bias
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**FunctionalValue**	**Comment**
	7:0	bias_vdac_val	08	RW	0x000000B1	0x000000B1	Voltage DAC value coded on 8 bits
	20:8	bias_idac_val	13	RW	0x00000345	0x00000345	Current DAC value coded on 13 bits
	26:21	*Reserved*	06	RW	0x00000008	0x00000008	
	27:27	bias_type	01	RW	0x00000000	0x00000000	Selects bias mode 0: current DAC 1: voltage DAC

	28:28	bias_mode	01	RW	0x00000001	0x00000001	Selects bias mode 0: cascode 1: normal
	29:29	*Reserved*	01	RW	0x00000000	0x00000000	
	30:30	*Reserved*	01	RW	0x00000001	0x00000001	
	31:31	*Reserved*	01	RW	0x00000000	0x00000000	

The photoreceptor bias is configured by default with the so-called IDAC mode which behavior is depicted in Figure 25. The IDAC has a log response versus the input code represented on 13 bits. Depending on the output voltage required, the output IDAC voltage range can be shifted down thanks to the "cascode feature". The normal mode is also selected by default.

The log fits are mentioned in Figure 25.

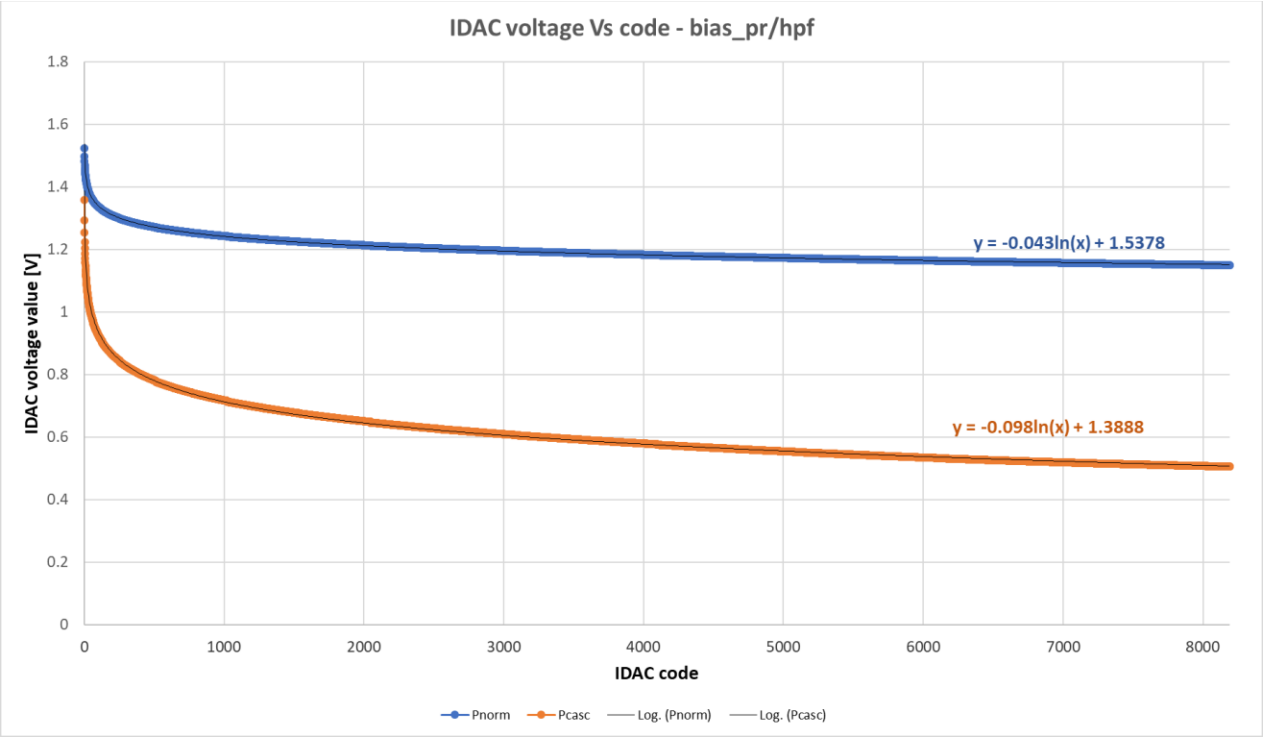


Figure 25: IDAC voltage values Vs code for bias_pr and bias_hpf.

8.9.2 Bias fo control

This bias controls the pixel low-pass cut-off frequency. It is the main bias. It is a tradeoff between pixel latency, pixel background noise.

Address		Register Name	ReadMask	WriteMask	ResetValue	FunctionalValue	Comment
0x00000158		bias_fo_ctrl	0xFFFFFFFF	0xFFFFFFFF	0x610036D6	0x610036D6	Pixel bandwidth bias
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**FunctionalValue**	**Comment**
	7:0	bias_vdac_val	08	RW	0x000000D6	0x000000D6	Voltage DAC value coded on 8 bits

	20: 8	bias_idac_val	13	RW	0x00000036	0x00000036	Current DAC value coded on 13 bits Railp DAC value coded on 7 bits [6:0]
	26:21	*Reserved*	06	RW	0x00000008	0x00000008	
	27:27	bias_type	01	RW	0x00000000	0x00000000	Selects bias mode 0: current DAC 1: voltage DAC
	28:28	bias_mode	01	RW	0x00000000	0x00000000	Selects bias mode 0: cascode 1: normal
	29:29	bias_fo_mode	01	RW	0x00000001	0x00000001	Selects bias_fo biasing mode. 0: normal. 1: railp.
	30:30	*Reserved*	01	RW	0x00000001	0x00000001	
	31:31	*Reserved*	01	RW	0x00000000	0x00000000	

The pixel low-pass bandwidth can be configured in 2 ways: the so-callepd IDAC mode which behavior is depicted in Figure 26 (voltage range is different from the one of bias_pr) and still on 13 bits. The other mode is the so-called RAILP which is coded on 7 bits which is shown in Figure 27. The IDAC has a log response versus the input code whereas the RAILP mode has a linear response. Depending on the output voltage required, the output IDAC voltage range can be shifted down thanks to the "cascode feature". The RAILP mode is selected by default. The RAILP mode provides a better control over temperature than the IDAC mode.

The log fits are mentioned in Figure 26.

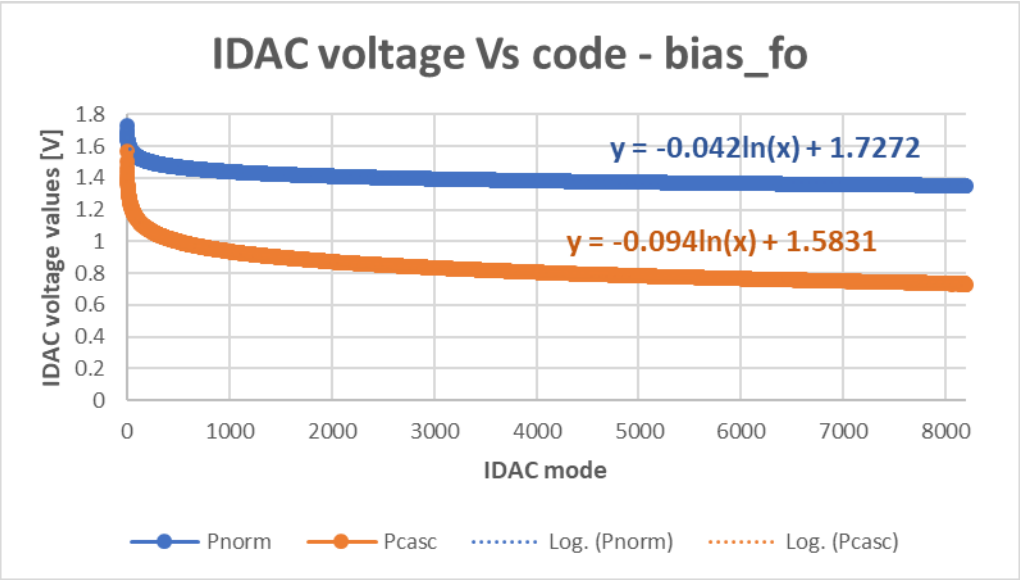


Figure 26: IDAC voltage values Vs code for bias_fo.

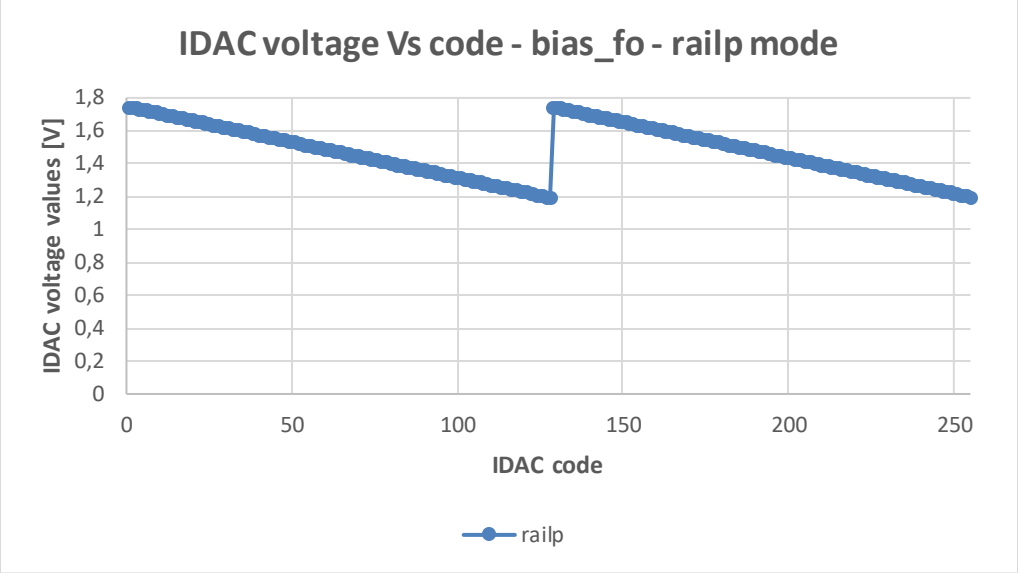


Figure 27: RAILP voltage values Vs code for bias_fo.

The linear fit is given in Figure 28.

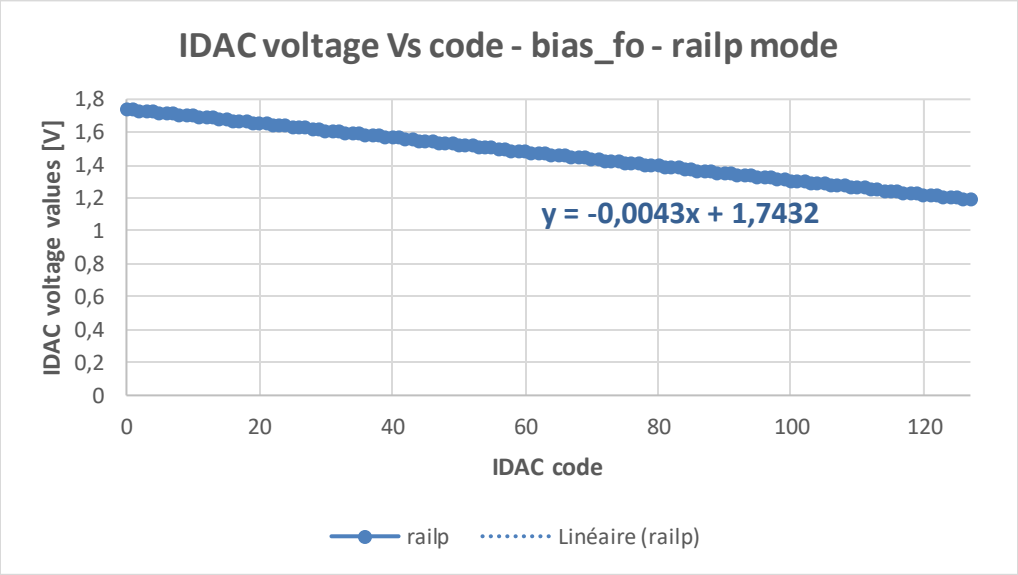


Figure 28: Linear fit of RAILP.

8.9.3 Bias diff control

This bias controls the reference voltage of the comparators. It is recommended to leave this value untouched.

Address		Register Name	ReadMask	WriteMask	ResetValue	FunctionalValue	Comment
0x00000154		bias_diff_ctrl	0xFFFFFFFF	0xFFFFFFFF	0x7102CE2A	0x7902CE29	Comparator reference voltage bias

	high:low	Field Name	size	Attribute	ResetValue	FunctionalValue	Comment
	7:0	bias_vdac_val	08	RW	0x0000002A	0x00000029	Voltage DAC value coded on 8 bits
	20:8	bias_idac_val	13	RW	0x000002CE	0x000002CE	Current DAC value coded on 13 bits
	26:21	*Reserved*	06	RW	0x00000008	0x00000008	
	27:27	bias_type	01	RW	0x00000001	0x00000000	Selects bias mode 0: current DAC 1: voltage DAC
	28:28	bias_mode	01	RW	0x00000001	0x00000001	Selects bias mode 0: cascode 1: normal
	29:29	*Reserved*	01	RW	0x00000001	0x00000001	
	30:30	*Reserved*	01	RW	0x00000001	0x00000001	
	31:31	*Reserved*	01	RW	0x00000000	0x00000000	

Bias diff, bias diff off, and bias_diff on are all set in VDAC mode by default in order to control properly the positive and negative contrast thresholds, especially in temperature. The VDAC output voltage has a linear function versus the input code and follows the following:

$$vdac\ voltage\ value = \frac{bias\ vdac\ val}{255} \cdot VDDA\ [V]$$

where VDDA is expressed in Volt.

8.9.4 Bias diff on controlp

This bias controls the positive comparator threshold. More precisely, the difference of bias_vdac_val between bias_diff and bias_diff_on defines the threshold voltage of the comparator related to ON events.

Address		Register Name	ReadMask	WriteMask	ResetValue	FunctionalValue	Comment
0x00000150		bias_diff_on_ctrl	0xFFFFFFFF	0xFFFFFFFF	0x71125C34	0x79125C33	Positive comparator threshold voltage bias
	high:low	Field Name	size	Attribute	ResetValue	FunctionalValue	Comment
	7:0	bias_vdac_val	08	RW	0x00000034	0x00000033	Voltage DAC value coded on 8 bits
	20:8	bias_idac_val	13	RW	0x0000125C	0x0000125C	Current DAC value coded on 13 bits
	26:21	*Reserved*	06	RW	0x00000008	0x00000008	
	27:27	bias_type	01	RW	0x00000001	0x00000000	Selects bias mode 0: current DAC 1: voltage DAC
	28:28	bias_mode	01	RW	0x00000001	0x00000001	Selects bias mode 0: cascode 1: normal
	29:29	*Reserved*	01	RW	0x00000001	0x00000001	
	30:30	*Reserved*	01	RW	0x00000001	0x00000001	
	31:31	*Reserved*	01	RW	0x00000000	0x00000000	

8.9.5 Bias diff off control

This bias controls the negative comparator threshold. More precisely, the difference of bias_vdac_val between bias_diff and bias_diff_off defines the threshold voltage of the comparator related to OFF events.

Address		Register Name	ReadMask	WriteMask	ResetValue	FunctionalValue	Comment
0x0000014C		bias_diff_off_ctrl	0xFFFFFFFF	0xFFFFFFFF	0x7100571F	0x7900571F	Negative comparator threshold voltage bias
	high:low	Field Name	size	Attribute	ResetValue	FunctionalValue	Comment
	7:0	bias_vdac_val	08	RW	0x0000001F	0x0000001F	Voltage DAC value coded on 8 bits
	20:8	bias_idac_val	13	RW	0x00000057	0x00000057	Current DAC value coded on 13 bits

	26:21	*Reserved*	06	RW	0x00000008	0x00000008	
	27:27	bias_type	01	RW	0x00000001	0x00000000	Selects bias mode 0: current DAC 1: voltage DAC
	28:28	bias_mode	01	RW	0x00000001	0x00000001	Selects bias mode 0: cascode 1: normal
	29:29	*Reserved*	01	RW	0x00000001	0x00000001	
	30:30	*Reserved*	01	RW	0x00000001	0x00000001	
	31:31	*Reserved*	01	RW	0x00000000	0x00000000	

8.9.6 Bias refr control

This bias controls the so-called refractory period. It represents a dead time for which the pixel will be kept in rest after an event is acknowledged.

Address		Register Name	ReadMask	WriteMask	ResetValue	FunctionalValue	Comment
0x00000160		bias_refr_ctrl	0xFFFFFFFF	0xFFFFFFFF	0x51004FD4	0x51004FD4	Refractory period bias
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**FunctionalValue**	**Comment**
	7: 0	bias_vdac_val	08	RW	0x000000D4	0x000000D4	Voltage DAC value coded on 8 bits
	20: 8	bias_idac_val	13	RW	0x0000004F	0x0000004F	Current DAC value coded on 13 bits
	26:21	*Reserved*	06	RW	0x00000008	0x00000008	
	27:27	bias_type	01	RW	0x00000000	0x00000000	Selects bias mode 0: current DAC 1: voltage DAC
	28:28	bias_mode	01	RW	0x00000001	0x00000001	Selects bias mode 0: cascode 1: normal
	29:29	*Reserved*	01	RW	0x00000000	0x00000000	
	30:30	*Reserved*	01	RW	0x00000001	0x00000001	
	31:31	*Reserved*	01	RW	0x00000000	0x00000000	

The bias is configured by default with the so-called IDAC mode which behavior is depicted in Figure 29. The IDAC has a log response versus the input code represented on 13 bits. Depending on the output voltage required, the output IDAC voltage range can be shifted down thanks to the “cascode fe ature”. The normal mode is also selected by default.

The log fits are mentioned in Figure 29.

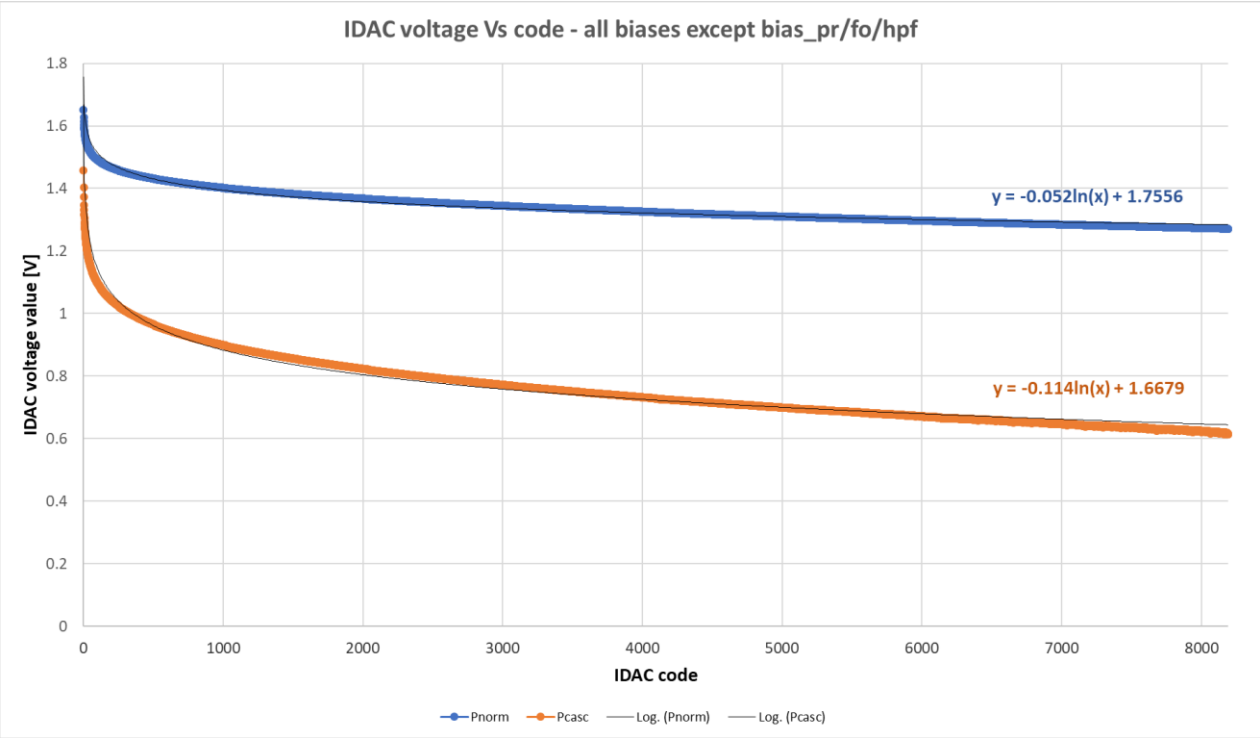


Figure 29: IDAC voltage values Vs code for all biases except bias_pr, bias_fo and bias_hpf.

8.9.7 Bias hpf control

This bias controls the high-pass filter function. It allows to reduce background noise. This bias is impacted by the value of bias_diff.

Address		Register Name	ReadMask	WriteMask	ResetValue	FunctionalValue	Comment
0x00000164		bias_hpf	0xFFFFFFFF	0xFFFFFFFF	0x510002D4	0x510002D4	Refractory period bias
	high:low	**Field Name**	**size**	**Attribute**	**ResetValue**	**FunctionalValue**	**Comment**
	7:0	bias_vdac_val	08	RW	0x000000D4	0x000000D4	Voltage DAC value coded on 8 bits
	20:8	bias_idac_val	13m	RW	0x0000004F	0x0000004F	Current DAC value coded on 13 bits
	26:21	*Reserved*	06	RW	0x00000008	0x00000008	
	27:27	bias_type	01	RW	0x00000000	0x00000000	Selects bias mode 0: current DAC 1: voltage DAC
	28:28	bias_mode	01	RW	0x00000001	0x00000001	Selects bias mode 0: cascode 1: normal
	29:29	*Reserved*	01	RW	0x00000000	0x00000000	
	30:30	*Reserved*	01	RW	0x00000001	0x00000001	
	31:31	*Reserved*	01	RW	0x00000000	0x00000000	

This bias uses the same bias unit than the one depicted in Figure 25.

8.10 Refractory clock

The refractory clock feature relies on a test pixel which continuously resets itself as soon as it is enabled via register tp_refr_clk_en as shown in the table below and Figure 30. The pseudo-clock generated from this

test pixel can be monitored on external pin REFRclk. The signal refr_clk represents twice the actual pixel refractory period.

Address		Register Name	ReadMask	WriteMask	ResetValue	FunctionalValue	Comment
0x0000001C		spare_ctrl	0xFFFFFFFF	0xFFFFFFFF	0x00000000	0x01000000	
	high:low	Field Name	size	Attribute	ResetValue	FunctionalValue	Comment
	24: 24	tp_refr_clk_en	1	RW	0x0	0x1	test pixel 0: refractory period clock is disabled 1: refractory period clock is enabled

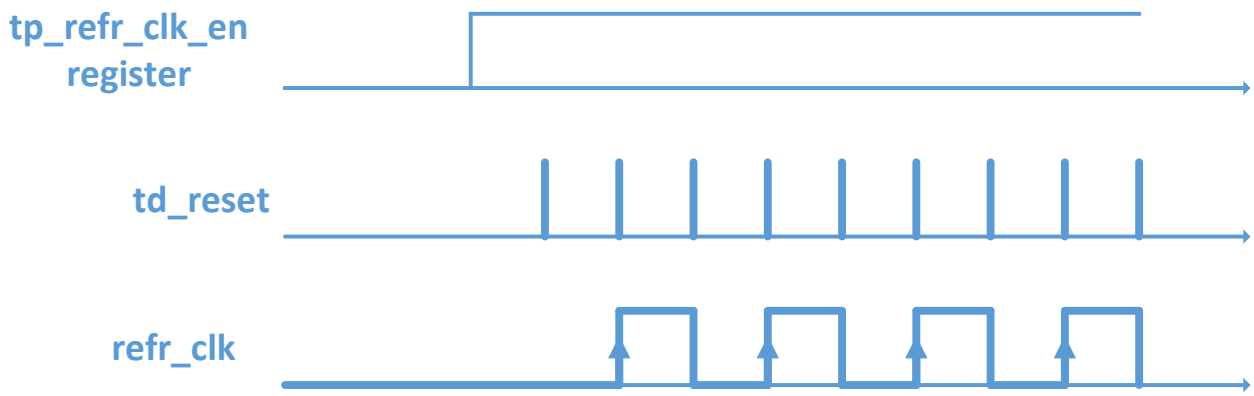


Figure 30: Refractory clock timing diagram.

The refractory period (or clock) varies function of voltage value of bias_refr as depicted in Figure 31. This figure already divides by 2 the refractory period.

From an application point of view, the refractory signal can be sampled on the external pin REFRclk with a digital counter running for instance at 100 MHz. The refractory period acquired by the digital counter will have to be divide by 2.

This feature can be continuously enabled or disabled on demand.

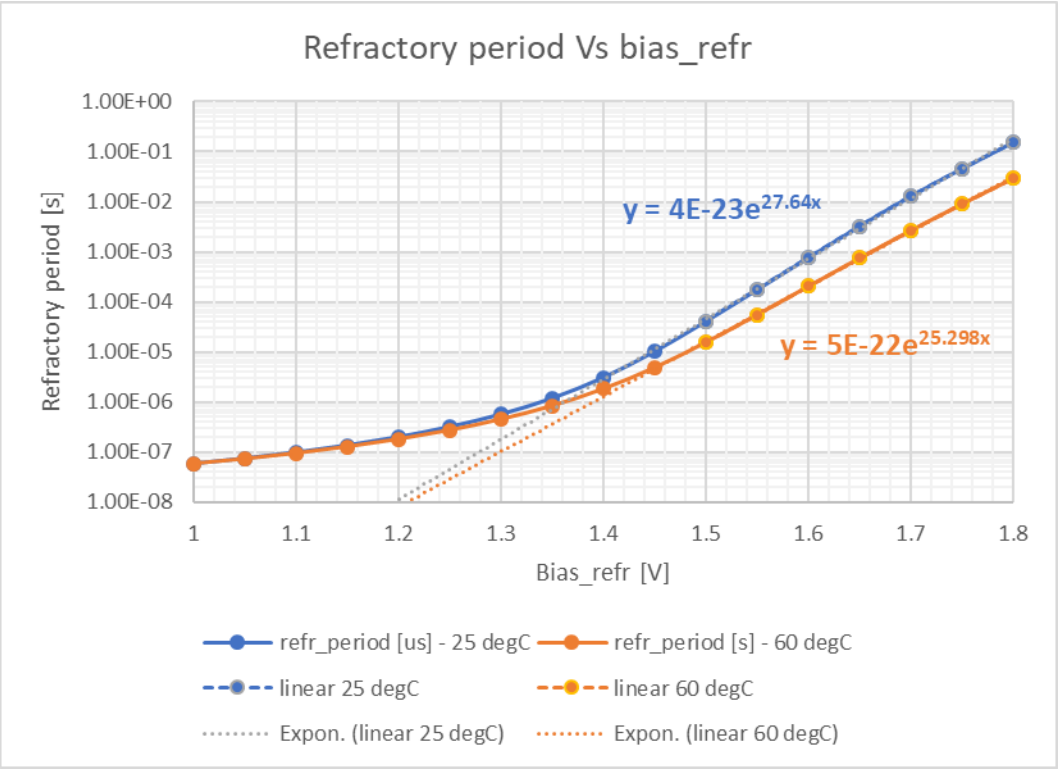
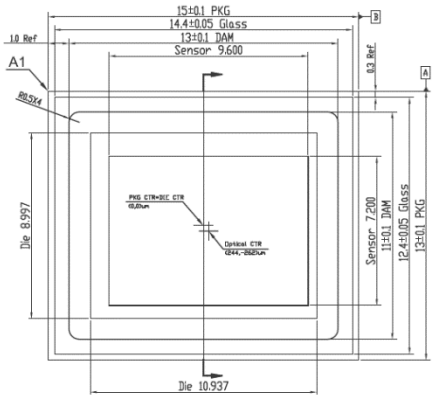


Figure 31: Refractory period function of bias refr voltage value.

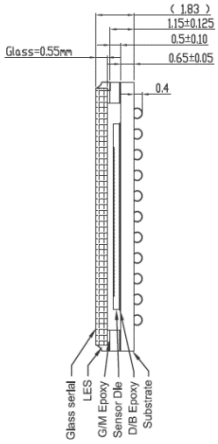
9 Assembly

9.1 Package

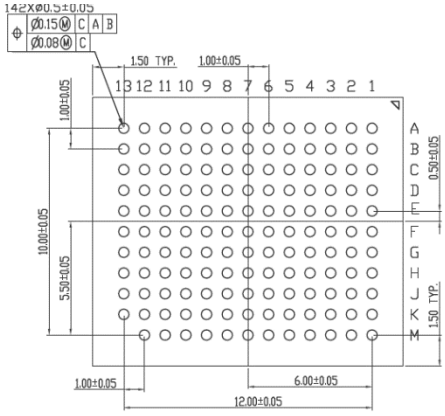
The product is offered in a mini PBGA 15x13mm 142 ball package.



TOP VIEW

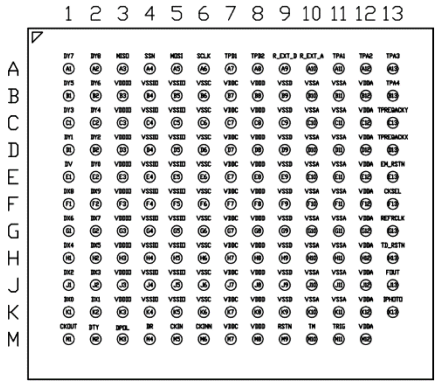


SIDE VIEW



BOTTOM VIEW

Die thickness = 300 um +/- 12um

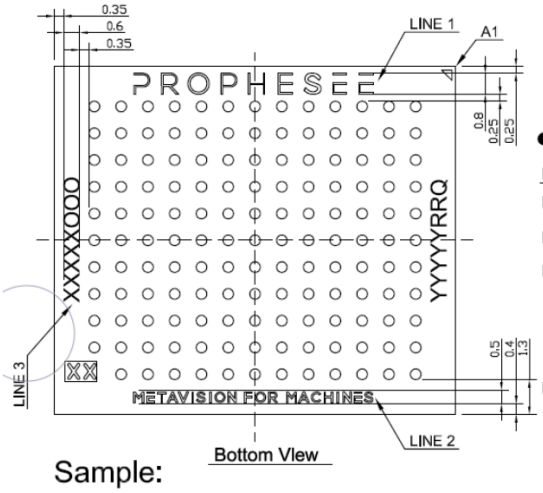


PIN ASSIGNMENT

Ball #	1	2	3	4	5	6	7	8	9	10	11	12	13
A	DY7	DY8	MISO	SSN	MOSI	SCLK	TPD1	TPD2	R_EXT_D	R_EXT_A	TPA1	TPA2	TPA3
B	DY5	DY6	VDDIO	VSSIO	VSSIO	VSSC	VDDC	VDDD	VSSD	VSSA	VSSA	VDDA	TPA4
C	DY3	DY4	VDDIO	VSSIO	VSSIO	VSSC	VDDC	VDDD	VSSD	VSSA	VSSA	VDDA	TPREQACKY
D	DY1	DY2	VDDIO	VSSIO	VSSIO	VSSC	VDDC	VDDD	VSSD	VSSA	VSSA	VDDA	TPREQACKX
E	DV	DY0	VDDIO	VSSIO	VSSIO	VSSC	VDDC	VDDD	VSSD	VSSA	VSSA	VDDA	EM_RSTN
F	DX8	DX9	VDDIO	VSSIO	VSSIO	VSSC	VDDC	VDDD	VSSD	VSSA	VSSA	VDDA	CKSEL
G	DX6	DX7	VDDIO	VSSIO	VSSIO	VSSC	VDDC	VDDD	VSSD	VSSA	VSSA	VDDA	REFRCLK
H	DX4	DX5	VDDIO	VSSIO	VSSIO	VSSC	VDDC	VDDD	VSSD	VSSA	VSSA	VDDA	TD_RSTN
J	DX2	DX3	VDDIO	VSSIO	VSSIO	VSSC	VDDC	VDDD	VSSD	VSSA	VSSA	VDDA	FOUT
K	DX0	DX1	VDDIO	VSSIO	VSSIO	VSSC	VDDC	VDDD	VSSD	VSSA	VSSA	VDDA	IPHOTO
M	CKOUT	DTY	DPOL	DR	CKIN	CKINN	VDDC	VDDD	RSTN	TM	TRIG	VDDA	

Bottom side (Top view see through)

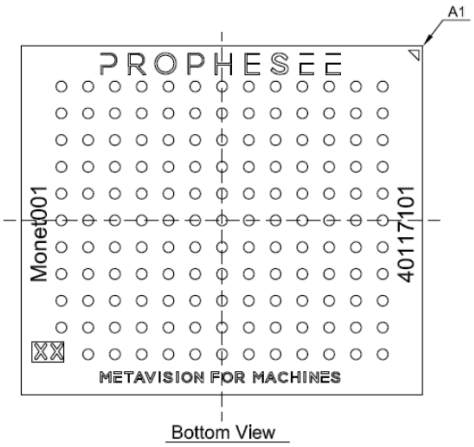
9.2 Package Markings



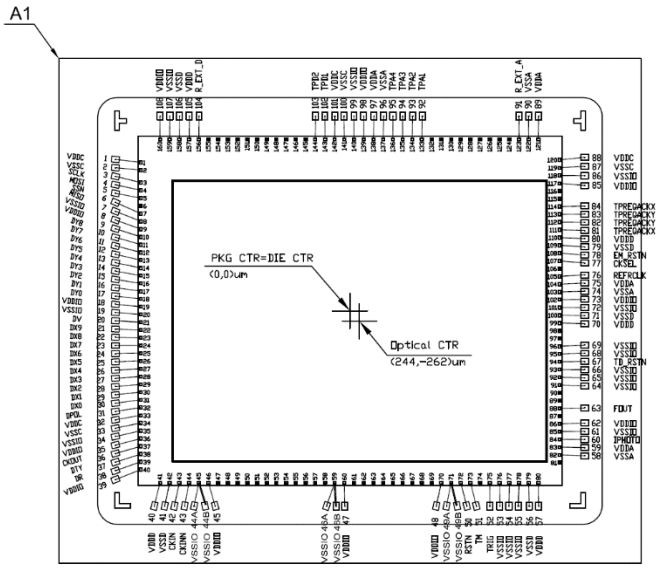
● Marking Explanation :

LINE#	說明 Explanation	內容 Contents	備註 Remark	字型 Font
LINE 1 :	LOGO	PROPHESEE		
LINE 2 :	LOGO	METAVISION FOR MACHINES		
LINE 3 :	Product Info & SBT ID	XXXXXOOO XXXXX:Product(Monet) OOO:Substrate ID(001~999) Example: Monet001		Arial
LINE 4 :	Sub lot numer	YYYYYRRQ YYYYY: sub lot number (E1940117.1.001) RR: sub lot number (E1940117.1.001) Q: sub lot number (E1940117.1.001) (Automatic data into the system)		Arial

Sample:



9.3 Optical center position



Pixel (0,0) is located in the bottom-left corner with respect to the drawing above.

9.4 Package Pinlist

Table 4: Package IO functionalities.

Pin	Pin Name	I/O	Power Domain	Description
A6	SCLK	I	VDDIO/VSSIO	Slave SPI clock
A5	MOSI	I	VDDIO/VSSIO	Slave SPI Master Output Slave Input
A4	CSn	I	VDDIO/VSSIO	Slave SPI chip select (active low)
A3	MISO	O	VDDIO/VSSIO	Slave SPI Master Input Slave Output
A2	DY8	O	VDDIO/VSSIO	Y-address bit 8
A1	DY7	O	VDDIO/VSSIO	Y-address bit 7
B2	DY6	O	VDDIO/VSSIO	Y-address bit 6
B1	DY5	O	VDDIO/VSSIO	Y-address bit 5
C2	DY4	O	VDDIO/VSSIO	Y-address bit 4
C1	DY3	O	VDDIO/VSSIO	Y-address bit 3
D2	DY2	O	VDDIO/VSSIO	Y-address bit 2
D1	DY1	O	VDDIO/VSSIO	Y-address bit 1
E2	DY0	O	VDDIO/VSSIO	Y-address bit 0
E1	DV	O	VDDIO/VSSIO	Data valid (active high)
F2	DX9	O	VDDIO/VSSIO	X-address bit 9
F1	DX8	O	VDDIO/VSSIO	X-address bit 8
G2	DX7	O	VDDIO/VSSIO	X-address bit 7
G1	DX6	O	VDDIO/VSSIO	X-address bit 6
H2	DX5	O	VDDIO/VSSIO	X-address bit 5
H1	DX4	O	VDDIO/VSSIO	X-address bit 4
J2	DX3	O	VDDIO/VSSIO	X-address bit 3
J1	DX2	O	VDDIO/VSSIO	X-address bit 2
K2	DX1	O	VDDIO/VSSIO	X-address bit 1
K1	DX0	O	VDDIO/VSSIO	X-address bit 0
M3	DPOL	O	VDDIO/VSSIO	X-address bit polarity
M1	CKOUT	O	VDDIO/VSSIO	Clock Out
M2	DTY	O	VDDIO/VSSIO	Leave pin floating
M4	DR	I	VDDIO/VSSIO	Data ready (active high)
M5	CKIN	I	VDDIO/VSSIO	Single-ended clock input.
M6	CKINn	I	VDDIO/VSSIO	Connect to ground.
M9	RSTn	I	VDDIO/VSSIO	Master Reset (active low)
M10	TM	I	VDDIO/VSSIO	Test Mode Enable (0: mode functional | 1: mode test)
M11	TRIG	I	VDDIO/VSSIO	Trigger Input. Deprecated, pull-down to ground.
K13	IPHOTO	IO	VDDA/VSSA	Leave pin floating.
J13	FOUT	O	VDDIO/VSSIO	PWM signal proportional to illuminance received on the sensor matrix. Default voltage is VDDA.
H13	TD_RSTn	I	VDDIO/VSSIO	Reset of the CD pixel matrix (active low)
G13	REFRCLK	O	VDDIO/VSSIO	Refractory clock
F13	CKSEL	I	VDDIO/VSSIO	0: Single-Ended |1: Differential Input Clock
E13	EM_RSTn	I	VDDIO/VSSIO	Deprecated connect to ground.
D13	TPREQACKX	O	VDDIO/VSSIO	Leave pin floating.
C13	TPREQACKY	O	VDDIO/VSSIO	Leave pin floating.
A10	r_ext_a	IO	VDDA/VSSA	Connect external resistor for analog bias generator
A11	TPA1	IO	VDDA/VSSA	Leave pin floating.
A12	TPA2	IO	VDDA/VSSA	Leave pin floating.
A13	TPA3	IO	VDDA/VSSA	Leave pin floating.
B13	TPA4	IO	VDDA/VSSA	Leave pin floating.
A7	TPD1	O	VDDIO/VSSIO	Leave pin floating.

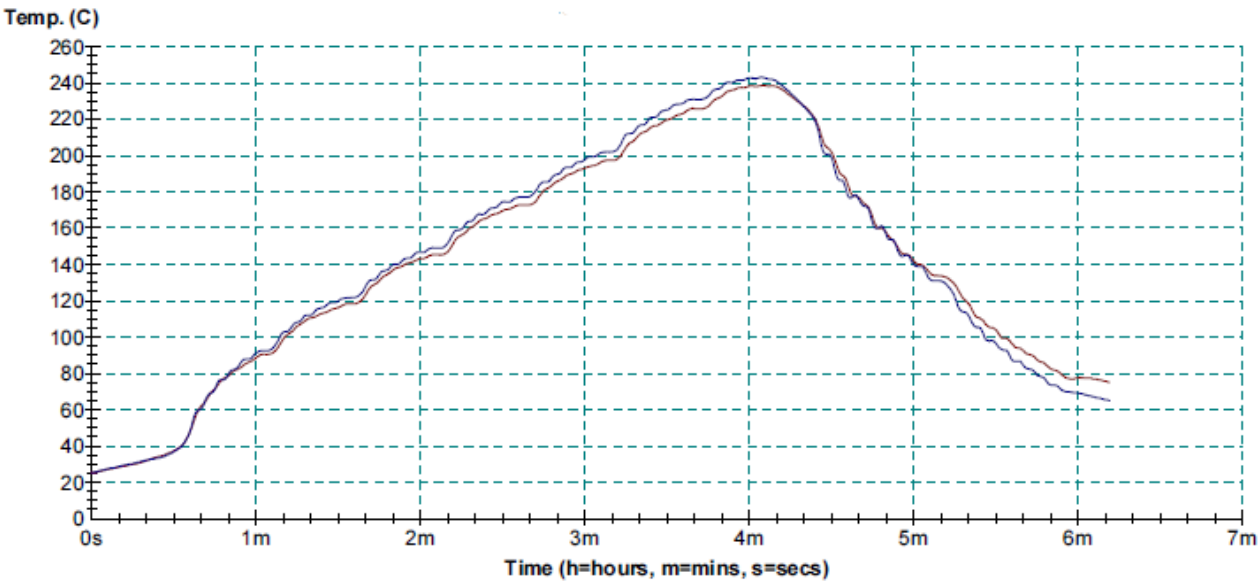
A8	TPD2	O	VDDIO/VSSIO	Leave pin floating.
A9	r_ext_d	IO	VDDIO/VSSIO	Connect external resistor for digital bias generator
B7,C7,D7,E7,F7,G7,H7,I7,J7,K7,M7	VDDC	-	VDDC/VSSC	Digital Core Supply
B6,C6,D6,E6,F6,G6,H6,I6,J6,K6	VSSC	-	VDDC/VSSC	Digital Core Ground
B3,C3,D3,E3,F3,G3,H3,I3,J3,K3	VDDIO	-	VDDIO/VSSIO	I/O Supply
B4,B5,C4,C5,D4,D5,E4,E5,F4,F5,G4,G5,H4,H5,I4,I5,J4,J5,K4,K5	VSSIO	-	VDDIO/VSSIO	I/O Ground
B8,C8,D8,E8,F8,G8,H8,I8,J8,K8,M8	VDDD	-	VDDD/VSSD	Asynchronous Logic Supply
B9,C9,D9,E9,F9,G9,H9,I9,J9,K9	VSSD	-	VDDD/VSSD	Asynchronous Logic Ground
B12,C12,D12,E12,F12,G12,H12,I12,J12,K12,M12	VDDA	-	VDDA/VSSA	Analog Core Supply
B10,B11,C10,C11,D10,D11,E10,E11,F10,F11,G10,G11,H10,H11,I10,I11,J10,J11,K10,K11	VSSA	-	VDDA/VSSA	Analog Core Ground

9.5 Glass Specification

Schott D 263® T Thin Glass

Technical Data	
Dimensions	440 mm x 360 mm, other size on request
Surface roughness	< 1 nm RMS
Thicknesses	0.03 mm up to 1.1 mm
Standard thicknesses and packaging units	0.21 mm 100 pcs 0.30 mm 100 pcs 0.40 mm 50 pcs 0.55 mm 50 pcs
Luminous transmittance τ_{vD65} (d = 1.1 mm)	91.7 %
Coefficient of mean linear thermal expansion α (20 °C; 300 °C) (static measurement)	7.2 x 10^{-6} K^{-1}
Transformation temperature Tg	557 °C
Dielectric constant ε_r at 1MHz	6.7
Refractive index n_D	1.5230
Refractive index n_e	1.5255 ± 0.0015
Density ρ (annealed at 40 °C/h)	2.51 g/cm³
Intensity of α-radiation	< 0.2 counts (h · cm²)*

9.6 Solder Temperature Profile



The two temperature profiles shown in the picture above are actual measurements. The profile does comply to the joint IPC/JEDEC standard IPC/J-STD-020D.1 section 5.6.

9.7 MSL: Moisture Sensitivity Level

Semiconductor components with resin-sealed package could be damaged during SMD reflow when moisture that got trapped inside the component expands.

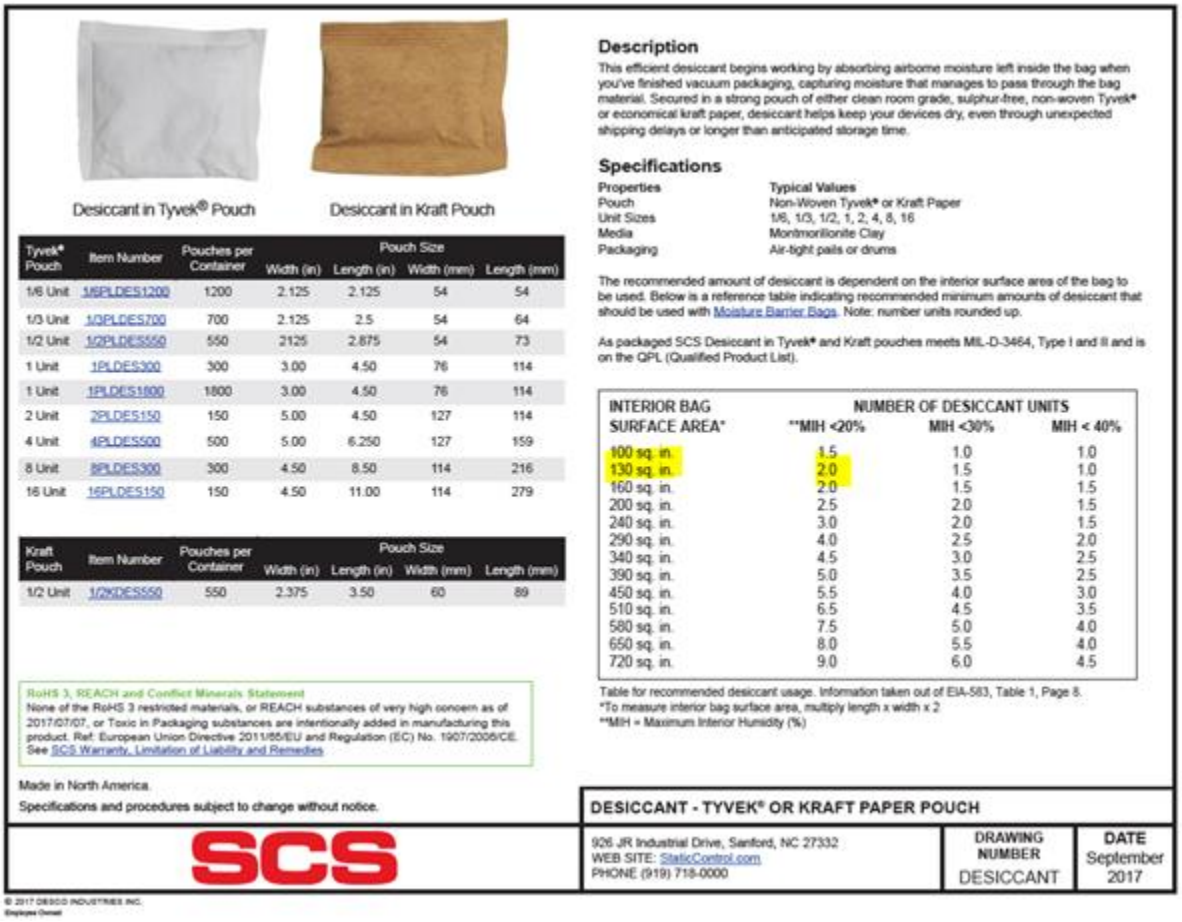
MSL is an electronic standard which is established by JEDEC for the time period in which a moisture sensitive device can be exposed to ambient room temperature. According to the standard, there are eight levels of

moisture sensitivity. The definition of Floor Life is the allowable period of time storage at less than 30°C, 60%RH.

MSL	Floor Life	Cond degC/%Rh
1	Unlimited (30°C less than 85%RH)	<=30/85%
2	1 year	<=30/60%
2a	4 weeks	<=30/60%
3	168 hours (7 days)	<=30/60%
4	72 hours (3 days)	<=30/60%
5	48 hours (2 days)	<=30/60%
5a	24 hours (1 day)	<=30/60%
6	Mandatory Bake before use	<=30/60%

The part PPS3MVCD is qualified to meet the **MSL-3 level** and passes the preconditioning tests according to JEDEC standard JESD22-A113.

Immediately following final test, the components shall be packed with 2 desiccant pouches per tray and the tray gets vacuum sealed.



Desiccant in Tyvek® Pouch

Desiccant in Kraft Pouch

Tyvek® Pouch	Item Number	Pouches per Container	Pouch Size Width (in)	Length (in)	Width (mm)	Length (mm)
1/8 Unit	1/8PLDES1200	1200	2.125	2.125	54	54
1/3 Unit	1/3PLDES700	700	2.125	2.5	54	64
1/2 Unit	1/2PLDES550	550	2125	2.875	54	73
1 Unit	1PLDES300	300	3.00	4.50	76	114
1 Unit	1PLDES1800	1800	3.00	4.50	76	114
2 Unit	2PLDES150	150	5.00	4.50	127	114
4 Unit	4PLDES500	500	5.00	6.250	127	159
8 Unit	8PLDES300	300	4.50	8.50	114	216
16 Unit	16PLDES150	150	4.50	11.00	114	279

Kraft Pouch	Item Number	Pouches per Container	Pouch Size Width (in)	Length (in)	Width (mm)	Length (mm)
1/2 Unit	1/2KDES550	550	2.375	3.50	60	89

RoHS 3, REACH and Conflict Minerals Statement
None of the RoHS 3 restricted materials, or REACH substances of very high concern as of 2017/07/07, or Toxic in Packaging substances are intentionally added in manufacturing this product. Ref: European Union Directive 2011/65/EU and Regulation (EC) No. 1907/2006/CE. See SCS Warranty, Limitation of Liability and Remedies

Made in North America.
Specifications and procedures subject to change without notice.

Description

This efficient desiccant begins working by absorbing airborne moisture left inside the bag when you've finished vacuum packaging, capturing moisture that manages to pass through the bag material. Secured in a strong pouch of either clean room grade, sulphur-free, non-woven Tyvek® or economical kraft paper, desiccant helps keep your devices dry, even through unexpected shipping delays or longer than anticipated storage time.

Specifications

Properties	Typical Values
Pouch	Non-Woven Tyvek® or Kraft Paper
Unit Sizes	1/8, 1/3, 1/2, 1, 2, 4, 8, 16
Media	Montmorillonite Clay
Packaging	Air-tight pails or drums

The recommended amount of desiccant is dependent on the interior surface area of the bag to be used. Below is a reference table indicating recommended minimum amounts of desiccant that should be used with Moisture Barrier Bags. Note: number units rounded up.

As packaged SCS Desiccant in Tyvek® and Kraft pouches meets MIL-D-3464, Type I and II and is on the QPL (Qualified Product List).

INTERIOR BAG SURFACE AREA*	NUMBER OF DESICCANT UNITS **MIH <20%	MIH <30%	MIH < 40%
100 sq. in.	1.5	1.0	1.0
130 sq. in.	2.0	1.5	1.0
160 sq. in.	2.0	1.5	1.5
200 sq. in.	2.5	2.0	1.5
240 sq. in.	3.0	2.0	1.5
290 sq. in.	4.0	2.5	2.0
340 sq. in.	4.5	3.0	2.5
390 sq. in.	5.0	3.5	2.5
450 sq. in.	5.5	4.0	3.0
510 sq. in.	6.5	4.5	3.5
580 sq. in.	7.5	5.0	4.0
650 sq. in.	8.0	5.5	4.0
720 sq. in.	9.0	6.0	4.5

Table for recommended desiccant usage. Information taken out of EIA-583, Table 1, Page 8.
*To measure interior bag surface area, multiply length x width x 2
**MIH = Maximum Interior Humidity (%)

DESICCANT - TYVEK® OR KRAFT PAPER POUCH

SCS

926 JR Industrial Drive, Sanford, NC 27332
WEB SITE: StaticControl.com
PHONE (919) 718-0000

DRAWING NUMBER: DESICCANT

DATE: September 2017

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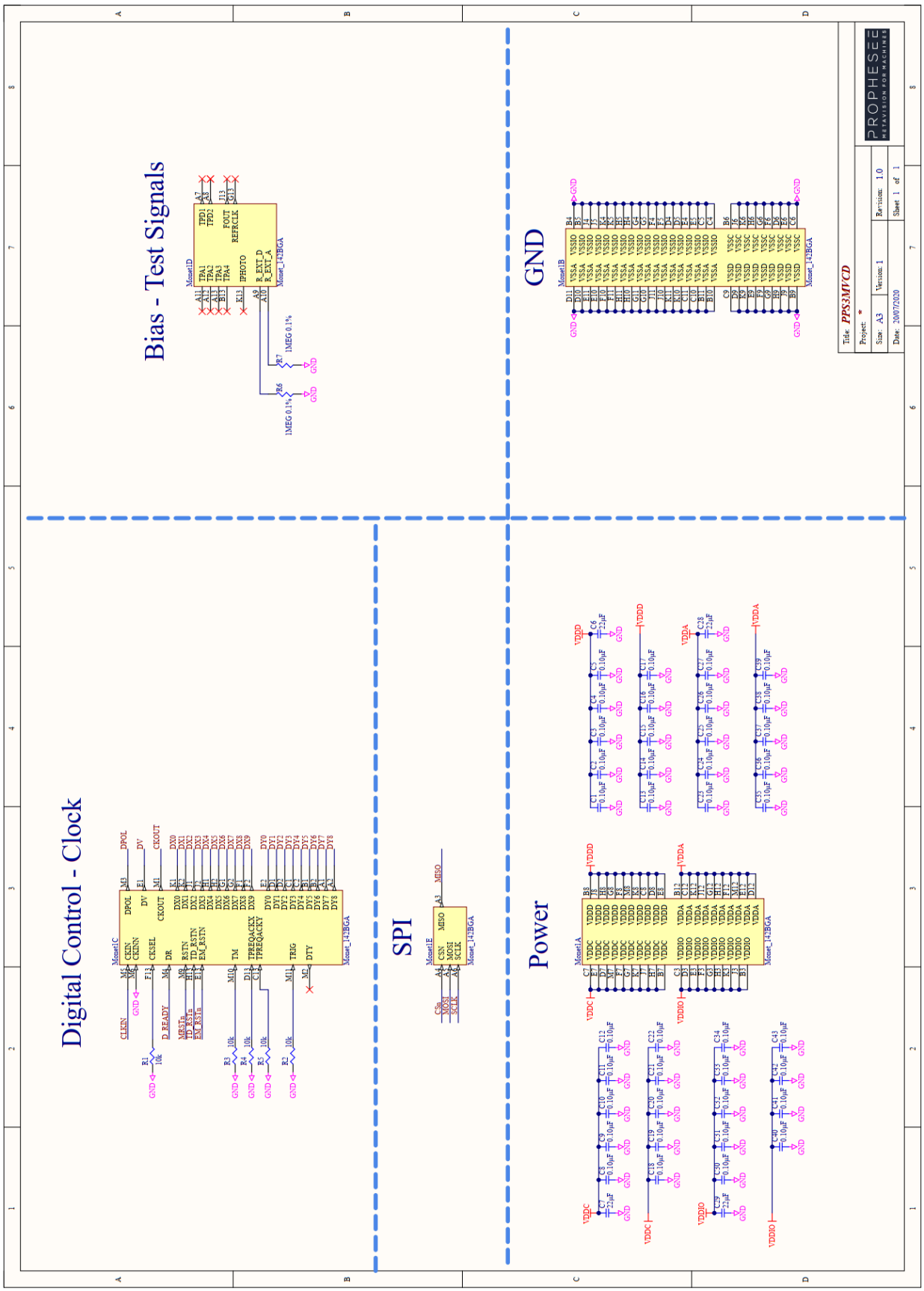
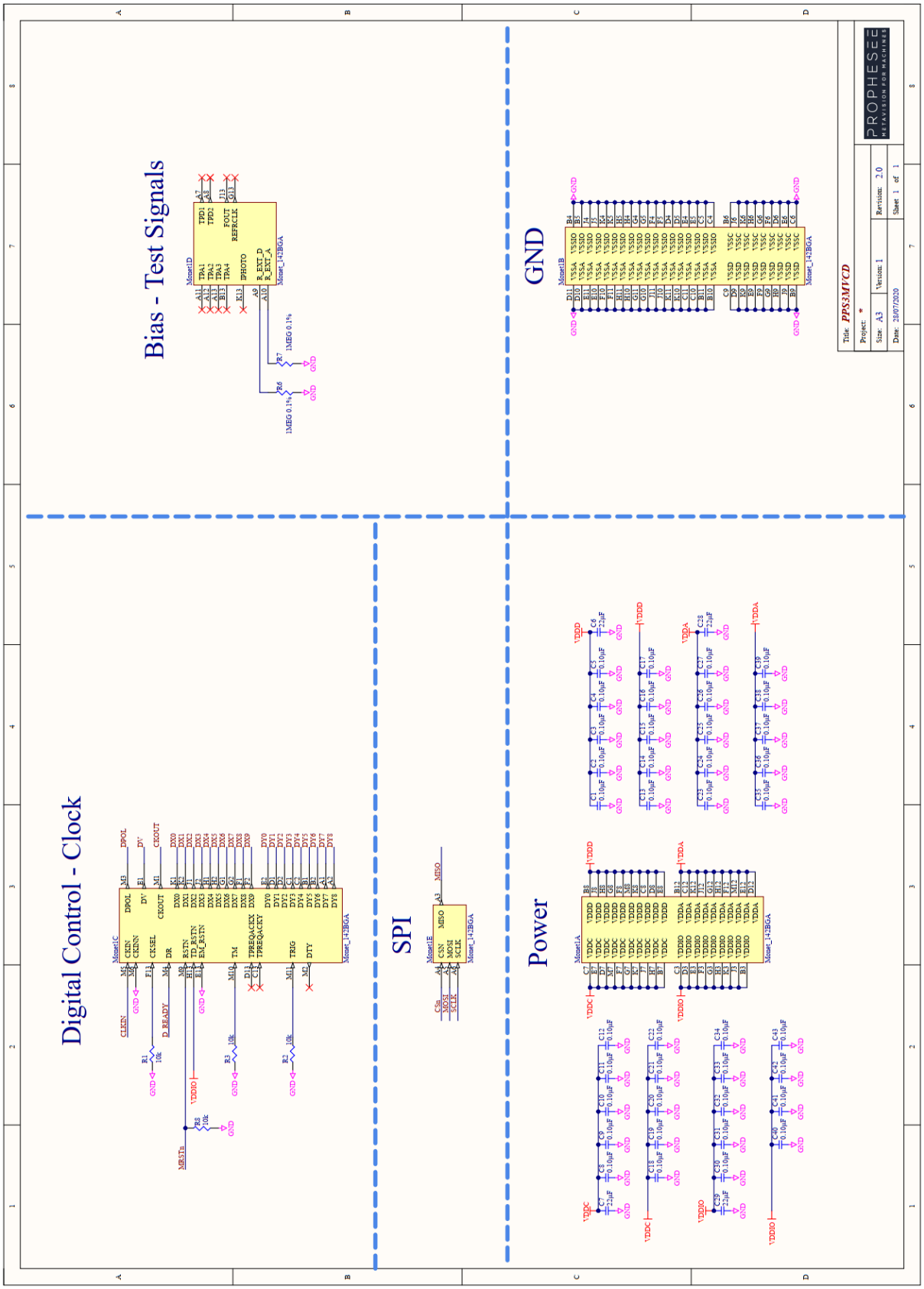
Employee Owned

10 Reference Schematic


Bias - Test Signals
GND
Digital Control - Clock
SPI
Power
Title: PPS3MVCD
Project: *
Size: A3
Version: 1
Revision: 1.0
Date: 20/07/2020
Sheet 1 of 1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Digital Control - Clock
SPI
Power
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Project: *
Size: A3
Version: 1
Revision: 2.0
Date: 28/07/2020
Sheet 1 of 1

10.1 Schematic Recommendations

10.1.1 General

Pull-down resistor should be added to RSTN signal to ensure the sensor remains in reset at start-up. Typically a value of 10KΩ is sufficient.

TD_RSTN can be tied to VDDIO to ensure the pixel matrix is always out of reset state. EM_RSTN can be connected to ground as its functionality is deprecated.

TM, TRIG, CKSEL and CKINN should be connected to ground during operation either directly or via a 10KΩ resistor.

Depending on the application a series termination resistor may be considered for each signal of the data interface. This value should be determined through board-level simulation to optimize signal integrity.

It is recommended to provide testpoints for TPAx, TPDx, TPREQACKX and TPREQACKY.

10.1.2 Power Supplies

All sensor power supply inputs are +1.8V, however care must be taken to ensure optimal performance.

Typically VDDIO and VDDC supplies can be shared by a single supply.

To provide optimal performance it is recommended to use a separate low-noise supply each for VDDA and VDDD. However VDDD may also be shared with VDDIO and VDDC in space-constrained applications. VDDA should always be given a low-noise supply and voltage variation on VDDA should be limited to not more than 5%.

Although separate ground domains are exposed at package level, it is recommended to use a common ground for all VSS pins.

10.2 Layout Recommendations

1. Place 0.1µF decoupling capacitors as close as possible between power and ground balls. This should ideally be directly beneath the balls on the reverse side of the board. Capacitor package size 0402 (01005) or smaller is preferred to minimize inductance.

2. Keep data interface traces as short as possible to optimize signal integrity. Ensure any length mismatch between traces is kept below 5mm.

3. Place 1MΩ ±0.1% resistors for REXTA and REXTD pins close to the sensor to minimize parasitic capacitance on these inputs and to minimize ground loop track. 0402 (01005) size packages preferred.

4. Provide dedicated power plane layers and a common ground plane layer.

5. All signals can be escaped on a single PCB layer to minimize board complexity.